L03 — study notes

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Lecture 3 (Topic 2B) — Taxes, bankruptcy costs and the static trade-off · Map of the notes

PDF. L03-notes.pdf — this whole unit in one printable file (rendered from these notes; rebuild with `python -I ~/.claude/commands/templates/notes_pdf.py Notas/L03` after any edit).

One claim. Taxes break MM through one door: interest is deducted, payouts to equity are not, so $V_L = V_U + V_{ITS}$ and the gain goes to shareholders. **What V_{ITS} is worth depends on the debt policy** — a fixed debt gives τD, a target ratio gives $\tau r_D D/(r_U - g)$ — and that choice is the whole difference between **APV and the after-tax WACC**, which agree exactly when D/E is constant. **Taken alone the shield says "all debt"; the deadweight cost of default says "not so fast"**, and the static trade-off puts the optimum where $\tau[1 - H(K^*)] = C\,h(K^*)$. **Every purely financial decision, payout included, stays a zero-NPV transaction** once these two frictions are accounted for.

Three layers per section:

Layer	What it means here
U — intuition	the two income statements, Higgins's Timid/Bold, the evidence on leverage, the investment-distortion numbers
C — course (examinable)	everything on slides 2–51 and 54–63: MM I with taxes, the balance-sheet identity and both MM II formulas, the three V_{ITS} and WACC forms, APV vs. WACC, the trade-off model and its FOC and statics, the bank-capital statements, MM dividend irrelevance, the appendix derivation
G — graduate	Miller (1977) personal taxes, DeAngelo–Masulis (Amaro §2.2). **Named, not examinable**

Reading order

Part I — the tax shield (slides 2–32, 54–63)

1. L03-01-tax-shield-mm-taxes — 2B.1–2B.2, slides 2–15; Berk §16.3 pp. 473–478, Ross §16.4 pp. 551–556, Amaro Prop. 2.1.3 p. 45. Settles $V_L = V_U + \tau D$ **derived line by line**, why

homemade leverage cannot copy it, the **two MM II formulas** (constant debt vs. target ratio), the **three V_{ITS} closed forms** — with the proof that slide 13's $1 + r_g$ is $1 + r_D$ — and the **repurchase at 0.8**.

2. L03-02-after-tax-wacc-apv — 2B.3–2B.4 and 2B.10, slides 16–32, 54–63; Berk pp. 476–478, Higgins pp. 321–322. Settles the **three WACC formulas from one identity**, the **APV = WACC proof**, the **project at 100,000 both ways and 98,945 with a fixed loan**, the **appendix derivation of the WACC**, and why a risky bond's shield uses the **coupon rate, not** r_D.

Part II — why not all debt (slides 33–45)

3. L03-03-debt-puzzle-bankruptcy-costs — 2B.5–2B.6, slides 33–40; Berk §16.4 pp. 479–480, Ross §16.5 pp. 556–558, Higgins pp. 208–213. Settles the **100%-debt puzzle and the evidence (12–44% debt)**, why **default itself is free in MM**, direct vs. indirect costs with magnitudes, and the **risk-shifting and debt-overhang numbers**.
4. L03-04-static-trade-off — 2B.7, slides 41–45; Berk §16.5 pp. 480–482, Ross §16.6 pp. 558–562. Settles $E(K)$**,** $D(K)$**,** $V(K)$ **derived**, the **FOC** $\tau[1-H] = Ch$, the **signs of** $dK^*/d\tau$ **and** dK^*/dC **by the implicit function theorem**, the two landmarks, and the **uniform case worked to** $K^* = 92.86$**,** $V^* = 81.63$ — plus the volatility caveat.

Part III — applications of the MM logic (slides 46–51)

5. L03-05-bank-capital-dividend-irrelevance — 2B.8–2B.9, slides 46–51; Amaro Prop. 4.1.1 p. 100, Berk §17.2 pp. 503–508, Ross §17.2 pp. 587–589. Settles **which bank-capital objections are MM fallacies and which is a subsidy**, and **dividend irrelevance from sources = uses**, with Wharton (173.55) and Genron (42) under every policy.

Support

6. L03-06-formulas — the block map as exam specification; every result with its assumption; every number.
7. L03-07-pitfalls — ten confusions with minimal counterexamples; 24 original True/False items.
8. L03-08-python — tax-shield, WACC, APV, a general-density trade-off solver, the dividend recursion; all self-checked.

The one result to hold

$$V_L = V_U + \underbrace{\frac{\tau\, E[\min(X, K)]}{1 + r_f}}_{V_{ITS}} - \underbrace{\frac{C\, H(K)}{1 + r_f}}_{PV(\text{bankruptcy costs})}, \qquad \tau\,[1 - H(K^*)] = C\, h(K^*)$$

With $C = 0$ it is MM I with taxes ($V_L = V_U + \tau D$, Amaro Prop. 2.1.3); with $\tau = 0$ it is MM (safe debt is irrelevant); with both it is the trade-off — **one formula, three lectures' worth of results.**

Numbers to have memorized

Case	Number	Meaning
Slide 2	ITS $= 350 \times 35\% = 122.5$	the shield is $\tau\times$ interest

Case	Number	Meaning
Slides 14–15	$V_L = 800$, $P = 0.6 \to 0.8$, gain 200	the whole V_{ITS} goes to the original shareholders
Slides 28–31	WACC 11.5%; $V_L = 100{,}000 = 93{,}750 + 6{,}250$	APV = WACC under a target ratio
Slide 32	$V_{ITS} = 1{,}200 \times 4.3295 = 5{,}195$; $V_L = 98{,}945$	a fixed schedule needs APV, at r_D
Slide 33	+7% at $D = 0.2V_U$, +17.5% at $0.5V_U$	the 100%-debt puzzle
Slide 34	debt 12% (high-tech) to 44% (durables) of value	nobody is near 100%
Note 04 calibration	$K^* = \bar{X} - C/\tau = 92.86$; $V^* = 81.63$	the trade-off optimum (declared calibration)

Figures

Built by `figures/make_figures.py` (run: `python -I Notas/L03/figures/make_figures.py`).

Figure	Shows	Lives in
L03-mm-taxes	$V_L = 600 + 0.4D$, equity $V_L - D$, share price $(V_U + \tau D)/1000$ (slides 14–15)	L03-01-tax-shield-mm-taxes §9
L03-its-policies	V_{ITS}/D vs. g under the three slide-13 policies	L03-01-tax-shield-mm-taxes §8
L03-wacc-taxes	r_E, r_U, after-tax WACC, r_D, $r_D(1-\tau)$ vs. D/V (slide 17 redrawn on slide-28 data)	L03-02-after-tax-wacc-apv §2
L03-apv-vs-wacc	100,000 / 100,000 / 98,945 and the two shield streams (slides 28–32)	L03-02-after-tax-wacc-apv §6
L03-trade-off	$V(K)$, the $C = 0$ curve, the distress wedge; marginal benefit vs. cost (slides 42–45)	L03-04-static-trade-off §5
L03-trade-off-statics	K^* vs. τ and vs. C, with the corners (slide 45)	L03-04-static-trade-off §5

Interactive companions

- The Trade-off Dial (`companions/companion-static-trade-off.html`) — slides 41–45 exactly (uniform h, tax only if solvent, fixed C), controls τ, C, $\underline{X}$, $\bar{X}$, r_f and an inspected K; read-outs K^* (the answer, with its regime), $V(K^*)$, the gain, PV shield, PV distress, default probability, D, E, D/V, promised yield; landmarks $C = 0$ and $\tau = 0$ one click away. Calibration declared in its footer; checked by `companions/check_companions.py`. Linked from L03-04-static-trade-off.

Scope and ceiling

- **Examinable:** slides 2–51 and the appendix 54–63, at the level of the C sections; the trade-off model with a general h (FOC, SOC, signs) and in the uniform case.
- **Named only:** the empirical papers on slides 34–35 and 38–39 (read their findings, not their methods); the bankruptcy procedure (Ross §16.10); Miller (1977) and DeAngelo–Masulis (G).
- **Out (scope guard):** dynamic trade-off and refinancing models, pecking order and signalling (Topic 6, asymmetric information), the formal agency models behind debt overhang and risk-shifting (Topic 4) — bridged to, not written up.

Course-wide position

Lecture	Relation to L03
L01 (Topic 1, accounting)	FCF and $EBIT(1-\tau)$ (L01 FCF); ROE rises with leverage — the bank-capital statement 1 (L01 ROE)
L02 (Topic 2A, valuation and MM)	the frictionless benchmark this lecture relaxes: MM I by arbitrage, risky debt as $\min(X, K)$ (L02 MM I), MM II and the pre-tax WACC (L02 MM II)
L03	**taxes, distress costs, the static trade-off, payout irrelevance**
Topic 4 (agency)	debt overhang, risk-shifting, cashing out as formal moral-hazard frictions (Tirole Ch. 3)
Topic 6 (asymmetric information)	why the static trade-off is not the whole story of financing choices

Rules pack: rules/03 (its "Taxes and default costs (trade-off)" section is still a placeholder; L03-06-formulas can fill it). Block map: lectures-index (Topic 2B). Index: Notas index.

Bibliography actually used

Source	Where
Ricca, Topic 2-B slides — the spine	every note
Berk, DeMarzo & Harford, *Fundamentals* 2e, §16.3–16.5 pp. 473–482; §17.2 pp. 503–508	Safeway table, Eqs. 16.4–16.10, Example 16.4, distress magnitudes, Genron (01–05)
Ross, Westerfield, Jordan & Lamb, *Fundamentos* 9e, §16.4–16.6 pp. 551–562; Table 16.7 p. 567; §16.10 pp. 567–568; §17.2 pp. 587–589	firms U and L, Eqs. 16.2–16.4, static theory, management advice, Wharton (01, 03–05)
Higgins, Koski & Mitton, 13e, Ch. 6 pp. 206–214; App. 6.A pp. 224–227; Ch. 8 App. pp. 319–322	Timid/Bold, distress costs, conflicts of interest, beta identity, APV (01–04)
Amaro de Matos, Ch. 2 §2.1.3–§2.2 pp. 44–58; Ch. 4 §4.1 pp. 99–101	Prop. 2.1.3, MM II with taxes, Miller, DeAngelo–Masulis, MM3 (01, 03–05)

Source	Where
Papers cited on the slides (Jagannathan et al. 2017; Graham et al. 2015; Hortaçsu et al. 2013; Brown & Matsa 2016; Baghai et al. 2021; Van Doornik et al. 2022; Admati et al. 2013; DeMarzo 2005; Stanton & Seasholes 2005)	as summarized on the slides (03, 05)

Related project material

- List: Lista R3 — MM, taxes and trade-off (15 exercises; numbers and pages in the notes' Links), its solutions (source `Listas/lista-r3-mm-taxes-tradeoff-solutions.tex`, check script `Listas/lista-r3-mm-taxes-tradeoff-solutions_check.py`), roadmap roteiro R3 and study pack.
- Problem set: PS2 Ex. 1(f) (p. 1), Ex. 2 and Ex. 3(c) (p. 2).
- Quiz: quiz Topic 2B (blocks `tax`, `dist`, `pol`).
- Audio prompt: `NotebookLM/audio/topic2b-audio.md`.

2B.1–2B.2 · The interest tax shield and MM I with corporate taxes

> **Claim.** Taxes do not break MM because they are taxes; they break it because **interest is deducted before tax and payouts to equity are not**. Each period the levered firm hands its investors $\tau r_D D$ more than the unlevered twin, so $V_L = V_U + V_{ITS}$, all of the gain goes to **shareholders**, and what V_{ITS} is worth depends entirely on **how risky the future debt level is** — a fixed debt gives τD, a target ratio gives $\tau r_D D/(r_U - g)$.

Prev: L03-00-map · Next: L03-02-after-tax-wacc-apv · Map: L03-00-map

Notation.

Symbol	Meaning	Range
τ	corporate (marginal) tax rate	$[0, 1)$
D, E	market values of debt and equity	≥ 0
r_D, r_f	expected return on debt; risk-free rate (safe debt: $r_D = r_f$)	> 0
r_E	expected return on levered equity	
r_U, β_U	expected return and beta of the **unlevered** firm's assets	$r_U > g$
X	operating cash flow (EBIT-like, pre-tax) at $t = 1$, growing at g	
V_U	value of the hypothetical unlevered firm	
V_{ITS}, r_{ITS}, β_{ITS}	PV of interest tax shields; its expected return and beta	
V_L	value of the levered firm $= D + E$	
ITS	interest tax shield of one period $= \tau \times$ interest	

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1 · U — two firms, one tax bill (slides 2–3)

The slide's two firms are Berk's Safeway 2008 (Table 16.2, printed p. 473): same assets, one with interest expense.

	Levered	Unlevered
EBITDA	2150	2150
Depreciation	300	300
EBIT	1850	1850
Interest expense	350	0
Pre-tax income	$1850 - 350 = 1500$	1850
Taxes (35%)	$0.35 \times 1500 = \mathbf{525}$	$0.35 \times 1850 = \mathbf{647.5}$
Net income	$1500 - 525 = 975$	$1850 - 647.5 = 1202.5$
Total paid to investors	$975 + 350 = \mathbf{1325}$	$\mathbf{1202.5}$

The levered firm's net income is lower, but what it pays **to all investors** is higher by

$$1325 - 1202.5 = 647.5 - 525 = 122.5 = 350 \times 0.35.$$

The gain is the tax not paid, and it equals the tax rate times the interest expense. With interest $= r_D \times D$:

$$ITS = \tau\, r_D\, D \tag{1}$$

(slide 3; Berk Eq. 16.4, p. 474, "Interest Tax Shield = Corporate Tax Rate × Interest Payments").

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2 · C — the unlevered FCF does not see the shield (slide 4)

Free cash flow (Topic 1, L01 §FCF) starts from $EBIT(1-\tau)$, which taxes EBIT **as if there were no interest**:

	Levered	Unlevered
$EBIT(1-35\%)$	$1850 \times 0.65 = 1202.5$	1202.5
Depreciation	300	300
FCF (no capex, no ΔNWC on the slide)	$1202.5 + 300 =$ **1502.5**	**1502.5**

FCF is identical across the two firms: it is the *unlevered* FCF, "as if the firm had no leverage". The 122.5 is missing from it by construction, so it must re-enter somewhere else — **either in the discount rate (after-tax WACC) or as a separate present value (V_{ITS}, APV)**. That choice is note L03-02-after-tax-wacc-apv. Counting it in both is double counting.

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3 · C — the friction is a *differential* treatment (slide 5)

With taxes, capital structure changes the cash flows that reach debt and equity holders. **A tax that hit interest and dividends alike would leave MM intact** (it would scale every claim by the same factor). The advantage of debt exists because, in most countries,

- interest payments are **deductible** from taxable income;
- dividends and retained earnings are **taxed** at the corporate level.

Higgins makes the same point from the other side (p. 226): where corporate and personal taxes are **fully integrated**, "there are no tax benefits to debt financing".

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4 · C — perpetual safe debt: $V_L = V_U + \tau D$ (slides 6–8)

Setup (slide 6). The firm generates X at $t = 1$, growing at g forever, and keeps a **constant** level of perpetual **safe** debt D, so interest is $r_f D$ each period.

Cash flow paid out each period = equity's after-tax cash flow + debt's interest:

$$(1-\tau)(X - r_f D) + r_f D \tag{2}$$

Expand the first product:

$$(1-\tau)X - (1-\tau)r_f D + r_f D \tag{3}$$

Collect the two $r_f D$ terms, $-(1-\tau)r_f D + r_f D = \tau r_f D$:

$$\underbrace{(1-\tau)X}_{\text{all-equity firm's after-tax cash flow}} + \underbrace{\tau r_f D}_{ITS} \tag{4}$$

Value the second term (slide 7). It is constant and as safe as the debt, so it is a perpetuity at r_f:

$$V_{ITS} = \frac{\tau r_f D}{r_f} = \tau D \tag{5}$$

Value the first term. It is the unlevered firm's cash flow, risky like the assets, growing at g: a growing perpetuity at r_U,

$$V_U = \frac{(1-\tau)X}{r_U - g} \tag{6}$$

Adding (5) and (6), the value of (4):

$$V_L = \frac{(1-\tau)X}{r_U - g} + \tau D \tag{7}$$

MM I with corporate taxes (slide 8). *The value of a levered firm equals that of an unlevered firm plus the present value of the interest tax shield:*

$$\boxed{V_L = V_U + V_{ITS}} \tag{8}$$

Same statement: Ross Prop. I with taxes, Eq. (16.3), p. 552; Berk Eq. (16.5), p. 475; Amaro de Matos **Proposition 2.1.3**, Eq. (2.7), p. 45 ($V_l^\tau = V_u^\tau + \tau B$).

Which assumption drives it. (5) needs the debt to be **permanent and safe** — that is what makes the shield a riskless perpetuity. Berk's Eq. (16.6)–(16.7) (p. 477) gets τD under a weaker condition: the principal is never repaid (rolled over or defaulted) and new debt is fairly priced, so $D = PV(\text{future interest})$ and $PV(ITS) = \tau\, PV(\text{future interest}) = \tau D$. **Amaro's version is the cleanest**: in his state-price model taxes are $\tau[V_j - F]^+$ (no tax in default), so equity + debt $= (1-\tau)V_j + \tau \min\{V_j, F\}$ state by state (p. 45), and summing with state prices gives $V_u^\tau + \tau B$ **even with risky debt** — the shield is τ times the market value of debt whatever that debt pays. That is exactly the $C = 0$ case of the trade-off model (L03-04-static-trade-off eq. (9)).

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5 · Worked — Ross's firms U and L (§16.4, pp. 551–553)

EBIT 1,000 per year forever for both; L has 1,000 of perpetual debt at 8%; $T_C = 30\%$; no depreciation, capex or ΔNWC.

	U	L
EBIT	1,000	1,000
Interest 0.08×1000	0	80
Taxable income	1,000	920
Taxes 30%	300	276
Cash flow to shareholders	700	$920 - 276 = 644$
Cash flow to creditors	0	80
Total	**700**	**724**

$$ITS = 80 \times 0.30 = 24, \qquad V_{ITS} = \frac{24}{0.08} = 300 = 0.30 \times 1000 \quad \text{(Ross 16.2)}$$

With $R_U = 10\%$: $V_U = 700/0.10 = 7{,}000$, and by (8) $V_L = 7{,}000 + 300 = 7{,}300$, so $E = 7{,}300 - 1{,}000 = 6{,}300$.

MM II for constant debt (Ross 16.4) — derived in §7 below as (11):

$$R_E = 10\% + (10\% - 8\%) \times \frac{1000}{6300} \times (1 - 0.30) = 10\% + 2\% \times 0.15873 \times 0.70 = 10\% + 0.2222\% = 10.22\%$$

$$\text{WACC} = \frac{6300}{7300}(10.2222\%) + \frac{1000}{7300}(8\%)(0.70) = 8.8219\% + 0.7671\% = 9.589\% \approx 9.6\%$$

Check with the constant-debt WACC of slide 18 (L03-02-after-tax-wacc-apv (8)): $10\% \times (1 - 0.30 \times 1000/7300) = 10\% \times 0.958904 = 9.589\%$. ✓

> Ross's own reading (p. 553): *"o VPL por real de dívida é de \$0,30. É difícil imaginar por que uma empresa não tomaria emprestado o máximo absoluto"* — the 100%-debt puzzle of L03-03-debt-puzzle-bankruptcy-costs.

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6 · C — who gets the gain, and why homemade leverage cannot copy it (slide 8)

Slide 8: *"If investors cannot get an equivalent tax break on homemade leverage, they are willing to pay a premium for levered firms. All the tax benefits from leverage accrue to the shareholders."*

Homemade leverage fails here — Higgins's Timid/Bold example (Table 6A.1, p. 225): both earn EBIT 400 on assets 1,000; Bold has 800 of debt at 10%; $\tau = 40\%$.

	Timid	Bold
EBIT − interest	$400 - 0 = 400$	$400 - 80 = 320$
Corporate tax 40%	160	128

	Timid	Bold
Earnings after tax	240	192
Investor's own outlay	200, borrowing 800 at 10% personally	200 (all of Bold's equity)
Investor's income	$240 - 80 = 160 \rightarrow$ **80%**	$192 \rightarrow$ **96%**

The personal loan's 80 of interest is **not** deducted at the 40% corporate rate, so the investor's replica earns 32 less: $192 - 160 = 32 = 0.40 \times 80$. **Without taxes the two rows coincide (160% each) and MM holds; with corporate taxes only the firm can buy the deduction.**

Why the shareholders get it. Creditors lend at a fair rate: the debt is worth D and they receive a fair return on D. Whatever is left of $V_L - D = V_U + V_{ITS} - D$ is equity — the whole V_{ITS} sits on the equity side (§9 makes this a share-price number).

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7 · C — the market-value balance sheet with taxes (slides 9–12)

Slide 9. The levered firm's assets are the unlevered assets **plus** the tax-shield asset:

Assets	Liabilities
V_U (r_U, β_U)	D (r_D, β_D)
V_{ITS} (r_{ITS}, β_{ITS})	E (r_E, β_E)
$V_L = V_U + V_{ITS}$	$= D + E$

Slide 10. A portfolio's return is the value-weighted return of what it holds; both sides hold the same total, so

$$\frac{D}{D+E} r_D + \frac{E}{D+E} r_E = \frac{V_U}{V_U + V_{ITS}} r_U + \frac{V_{ITS}}{V_U + V_{ITS}} r_{ITS} \tag{9}$$

$$\frac{D}{D+E} \beta_D + \frac{E}{D+E} \beta_E = \frac{V_U}{V_U + V_{ITS}} \beta_U + \frac{V_{ITS}}{V_U + V_{ITS}} \beta_{ITS} \tag{10}$$

Higgins writes (10) in his Ch. 8 appendix footnote 10 (p. 319) from $D + E = V_u + tD$.

Everything now hinges on r_{ITS} — how risky is the shield? Two policies.

7.1 · Constant debt level: $r_{ITS} = r_D$ (slide 11)

If the firm keeps a constant market value of debt forever, the shield is as risky as the debt, and (5) with r_D in place of r_f gives $V_{ITS} = \tau D r_D / r_D = \tau D$. Multiply (9) through by $V_L = D + E = V_U + \tau D$:

$$D r_D + E r_E = V_U r_U + \tau D \, r_D$$

Substitute $V_U = D + E - \tau D$:

$$D r_D + E r_E = (D + E - \tau D) \, r_U + \tau D \, r_D$$

Move Dr_D to the right and group by D:

$$Er_E = Er_U + D(1-\tau)r_U - D(1-\tau)r_D$$

Divide by E:

$$\boxed{r_E = r_U + \frac{D}{E}(1-\tau)(r_U - r_D)} \qquad \text{(constant debt)} \tag{11}$$

— Ross Eq. (16.4), p. 553; Amaro p. 46 ($R_s^\tau = R_s^u + (R_s^u - R)\frac{B}{S^\tau}(1-\tau)$). The same steps on (10) give $\beta_E = \beta_U + \frac{D}{E}(1-\tau)(\beta_U - \beta_D)$; with safe debt ($\beta_D = 0$), **Hamada's** $\beta_E = \beta_U[1 + (1-\tau)D/E]$ (Amaro p. 46).

7.2 · Constant target D/E: $r_{ITS} = r_U$ (slide 12)

If the firm **continuously adjusts** its debt to keep D/E constant, debt rises and falls with the assets, and so do the shields: $\beta_{ITS} = \beta_U$, $r_{ITS} = r_U$. The right side of (9) becomes $\frac{V_U}{V_L}r_U + \frac{V_{ITS}}{V_L}r_U = r_U$:

$$\frac{D}{D+E}r_D + \frac{E}{D+E}r_E = r_U \tag{12}$$

Multiply by $D+E$, subtract Dr_D, divide by E, using $\frac{D+E}{E} = 1 + \frac{D}{E}$:

$$r_E = \Big(1 + \frac{D}{E}\Big)r_U - \frac{D}{E}r_D$$

$$\boxed{r_E = r_U + \frac{D}{E}(r_U - r_D)} \qquad \text{(target } D/E\text{)} \tag{13}$$

— **the no-tax MM II formula survives unchanged**, with r_U in the role of r_A (L02 §MM II). Likewise $\beta_U = \frac{D}{V}\beta_D + \frac{E}{V}\beta_E$; with $\beta_D = 0$, $\beta_U = \frac{E}{V}\beta_E$ — Higgins's unlevering formula $\beta_A = \frac{E}{V}\beta_E$ (p. 319), which he obtains from (10) by assuming exactly $\beta_{ITS} = \beta_A$.

> **Slide 12: "This is the usual assumption in valuation."** Unlevering comparables with $\beta_U = \frac{E}{V}\beta_E + \frac{D}{V}\beta_D$ (Topic 2A) is *already* the target-ratio assumption. Using Hamada's $(1-\tau)$ version silently assumes the opposite policy.

---

8 · C — three closed forms for V_{ITS} when FCF grows at g (slide 13)

Let D be today's debt and $r_U > g$. A growing perpetuity paying $c(1+g)^{t-1}$ at $t = 1, 2, ...$ is worth $\sum_{t\geq 1} c(1+g)^{t-1}/(1+r)^t = c/(r-g)$.

(a) Constant debt level. $ITS_t = \tau r_D D$ every period, discounted at r_D:

$$V_{ITS} = \sum_{t\geq 1} \frac{\tau r_D D}{(1+r_D)^t} = \frac{\tau r_D D}{r_D} = \tau D \tag{14}$$

(b) Target D/E, continuously adjusted. Expected debt grows with the firm, $E_0[D_t] = D(1+g)^t$; $ITS_{t+1} = \tau r_D D_t$ carries asset risk all the way, so it is discounted at r_U:

$$V_{ITS} = \sum_{t\geq 1} \frac{\tau r_D D(1+g)^{t-1}}{(1+r_U)^t} = \frac{\tau r_D D}{r_U - g} \tag{15}$$

(c) Target D/E, adjusted once a year. Debt D_t is set at date t and held for a year, so $ITS_{t+1} = \tau r_D D_t$ is **known at** t: worth $\tau r_D D_t/(1+r_D)$ at t. Only D_t itself is risky (like the assets) between 0 and t:

$$V_{ITS} = \sum_{t\geq 0} \frac{1}{(1+r_U)^t} \cdot \frac{\tau r_D D(1+g)^t}{1+r_D} = \frac{\tau r_D D}{1+r_D} \sum_{t\geq 0} \Big(\frac{1+g}{1+r_U}\Big)^t$$

The geometric sum is $1/(1-\frac{1+g}{1+r_U}) = \frac{1+r_U}{r_U-g}$, so

$$\boxed{V_{ITS} = \frac{\tau r_D D}{r_U - g} \cdot \frac{1+r_U}{1+r_D}} \tag{16}$$

> **Slide 13 prints the second factor as $\frac{1+r_U}{1+r_g}$; derivation (16) shows the denominator must be $1+r_D$** — the one period over which the shield is as safe as the debt. The same correction applies to the annually adjusted WACC on slide 18. (16) lies between (15) and a version discounted wholly at r_D, and since $r_U > r_D$, (16) > (15).

At the slide-28 project's numbers ($D = 25{,}000$, $r_D = 5\%$, $\tau = 40\%$, $r_U = 12\%$, $g = 4\%$):

Policy	Formula	V_{ITS}
(b) continuous	$\frac{0.40 \times 0.05 \times 25{,}000}{0.12 - 0.04} = \frac{500}{0.08}$	**6,250** (the slide-31 value)
(c) annual	$6{,}250 \times \frac{1.12}{1.05} = 6{,}250 \times 1.066667$	6,666.67
(a) fixed D, no growth of debt	$0.40 \times 25{,}000$	10,000

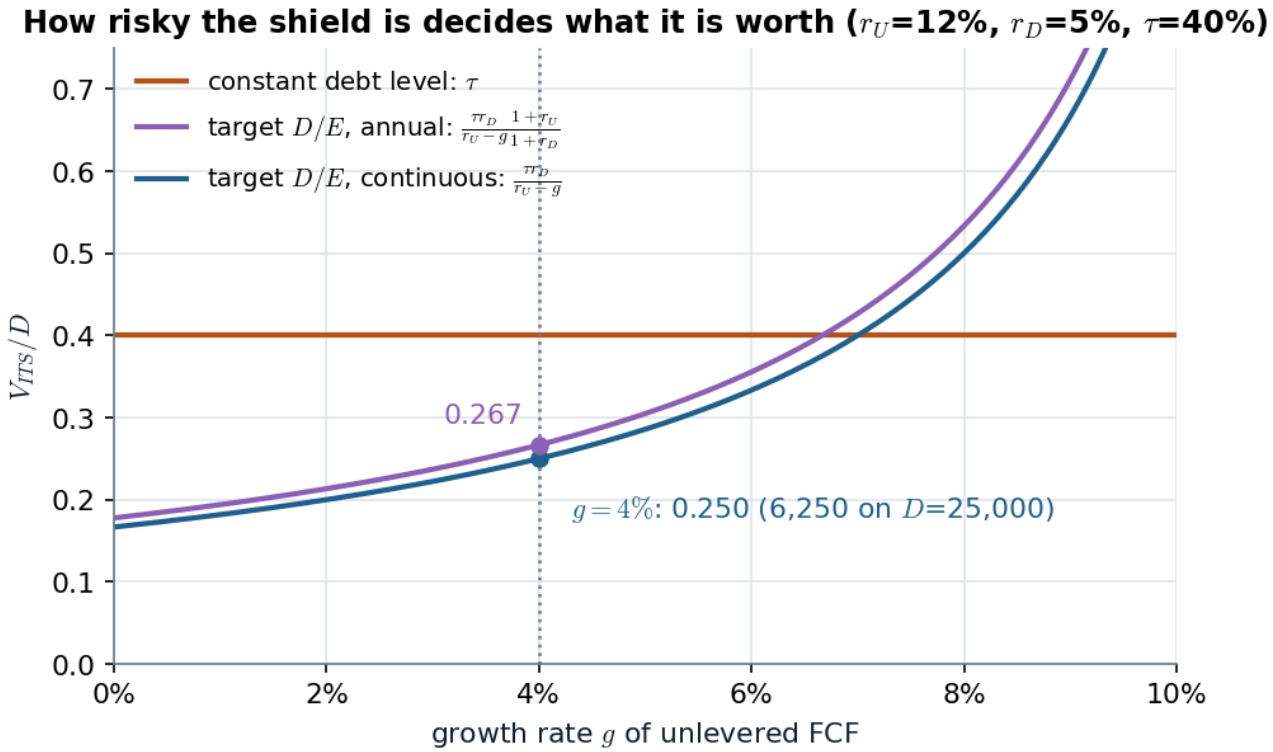


Figure — V_{ITS}/D under slide 13's three policies at $r_U = 12\%$, $r_D = 5\%$, $\tau = 40\%$: constant debt is flat at $\tau = 0.40$; a target ratio gives 0.250 (continuous) and 0.267 (annual) at $g = 4\%$. Source: eqs. (14)–(16), slide 13.

A fixed debt level makes the shield safe, so it is worth more per unit of today's debt than a target-ratio shield at low growth; at high g the target-ratio shield wins because the debt behind it grows with the firm. **The policy, not the tax rate, sets the discount rate of the shield.**

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9 · Worked — the debt-financed repurchase (slides 14–15)

Data. Unlevered firm worth 600, 1,000 shares → $P_0 = 0.6$. It borrows 500 of perpetual safe debt and uses it to repurchase shares. $\tau = 40\%$.

Value after the announcement, by (8) and (5):

$$V_L = 600 + 0.40 \times 500 = 600 + 200 = 800 \tag{17}$$

Equity, after the 500 of debt is in place:

$$E = V_L - D = 800 - 500 = 300 \tag{18}$$

The repurchase price. With competition and perfect information the repurchase is a zero-NPV trade: sellers must be indifferent between selling and keeping, so they sell at the post- announcement price P. With N shares bought,

$$P \times N = 500 \qquad \text{(the amount raised)} \tag{19}$$

$$P \times (1000 - N) = 300 \qquad \text{(equity after the buyback)} \tag{20}$$

Add (19) and (20): $P \times 1000 = 800$, so

$$P = 0.8, \qquad N = \frac{500}{0.8} = 625, \qquad 1000 - 625 = 375 \text{ shares left}, \qquad 0.8 \times 375 = 300 \ \checkmark$$

The gain. Every original shareholder — those who sell and those who stay — gains $0.8 - 0.6 = 0.2$ per share:

$$0.2 \times 1000 = 200 = V_{ITS}$$

The price jumps at the announcement, not at the repurchase. Shares are bought back at 0.8, already incorporating the shield; that is why the sellers also capture it. Lista R3 Ex. 12 (Berk 16.18, printed p. 493) and PS2 Ex. 2(c), 2(e) (`Listas/PS2.pdf`, p. 2) drill exactly this sequence.

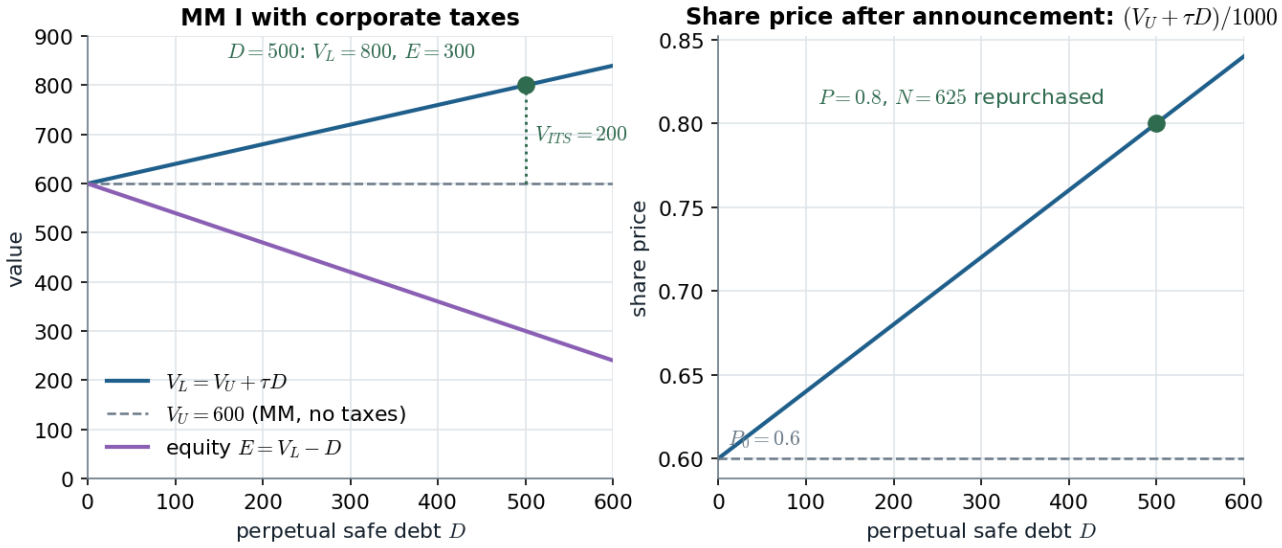


Figure — slide 14's firm: V_L rises with slope $\tau = 0.4$ over the no-tax MM line; equity falls by only $(1-\tau)$ per unit of debt; the price after announcement is $(V_U + \tau D)/1000$, 0.8 at $D = 500$. Source: eqs. (17)–(20), slides 14–15.

The purple line is the whole story of who wins: equity falls by D (the claim handed to creditors) but rises by τD (the shield), so each unit of debt-financed buyback costs the remaining equity only $1 - \tau$ of value while paying sellers full price.

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10 · G — personal taxes: Miller (1977) (not in the slides; named, not examinable)

Amaro de Matos §2.2 (pp. 48–54) adds personal tax rates on bond income τ_b and on equity income τ_s. Miller's equilibrium in the bond market sets (Eq. 2.11, p. 51)

$$(1 - \tau_s)(1 - \tau_c) = 1 - \tau_b(B^*)$$

and the gain from leverage for stockholders (p. 52) is $G_L = \left[1 - \frac{(1-\tau_s)(1-\tau_c)}{1-\tau_b(B^*)}\right]\frac{r_\circ K}{1-\tau_s}$, which (2.11) makes **zero**: in equilibrium no single firm gains from debt, though aggregate debt is determinate. With $\tau_s = \tau_b = 0$ the gain is back to τ_c per unit of debt — this note's (5). (Lista R3 Ex. 14, Amaro 2.2.7 #1, printed p. 58, asks for the general G_L; not reproduced here.) Higgins's lower panel of Table 6A.1 (p. 225–226) shows the milder version: a 33% personal tax on **both** firms' income cuts Bold's edge from 32 to $129 - 107 = 22$ but does not remove it.

Links

- Next: L03-02-after-tax-wacc-apv (where the shield enters the discount rate); L03-04-static-trade-off (the shield when debt is risky and default costly).
- Before: L02 MM I (homemade leverage, the no-tax benchmark), L02 MM II; FCF definition L01 FCF.
- Rules: rules/03. Block map: lectures-index (Topic 2B, 2B.1–2B.2).
- Practice: Lista R3 Ex. 3 (Ross 16.14, p. 578), Ex. 9 (Grinblatt–Titman 13.15–13.16, p. 494), Ex. 12 (Berk 16.18, p. 493), Ex. 14 (Amaro 2.2.7 #1, p. 58); PS2 Ex. 2(a)–(c), (e) (PS2, p. 2); quiz block `tax` of quiz Topic 2B.
- Code: L03-08-python §1.

2B.3–2B.4, 2B.10 · After-tax WACC, APV, and why they agree only under a target ratio

> **Claim.** There are two ways to put the shield back into a value built on unlevered FCF: **discount the FCF at a lower rate** (the after-tax WACC, r_D replaced by $r_D(1-\tau)$), or **value the shield separately** (APV: V_U at r_U plus V_{ITS} at r_{ITS}). They give the **same number exactly when D/V is held constant** — the slide project is worth 100,000 both ways — and **APV alone survives a debt schedule that is fixed in currency** (the 60,000 loan: 98,945). The WACC formula is not an assumption: the appendix **derives** it from $V_L = V_U + V_{ITS}$.

Prev: L03-01-tax-shield-mm-taxes · Next: L03-03-debt-puzzle-bankruptcy-costs · Map: L03-00-map

Notation. As in L03-01-tax-shield-mm-taxes, plus: FCF_t unlevered free cash flow of period t; r_{WACC} after-tax WACC; r_L expected return on the levered firm's assets ($V_U + V_{ITS}$); $C_{L,t+1}$ cash flow to all investors; $\bar{r}_D$ the coupon (interest) rate that sets the tax deduction; C_{t+1} the bond's coupon; $D_t, E_t, V_{L,t}$ values at t.

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1 · C — the after-tax WACC (slide 16)

Interest is deductible, so a unit of interest costs the firm only $1-\tau$ after tax: the **effective** cost of debt is $r_D(1-\tau)$. If the firm keeps a constant target D/E (continuously adjusted),

$$\boxed{r_{WACC} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D(1-\tau)} \tag{1}$$

and **discounting the unlevered FCF at (1) gives the value of the levered firm** — Berk Eq. (16.8), p. 477. Berk rewrites it (Eq. 16.9) as the pre-tax WACC minus the shield:

$$r_{WACC} = \underbrace{\frac{E}{E+D} r_E + \frac{D}{E+D} r_D}_{\text{pre-tax WACC}} - \underbrace{\frac{D}{E+D}\tau r_D}_{\text{reduction due to the ITS}} \tag{2}$$

2 · C — three WACC formulas (slide 18), derived

For an FCF growing at g forever, $V_L = FCF_1/(r_{WACC} - g)$, and also $V_U = FCF_1/(r_U - g)$. Divide:

$$r_{WACC} - g = \frac{FCF_1}{V_L} = (r_U - g)\frac{V_U}{V_L} = (r_U - g)\Big(1 - \frac{V_{ITS}}{V_L}\Big) \tag{3}$$

so **every WACC formula is (3) with one of slide 13's V_{ITS}** (L03-01-tax-shield-mm-taxes (14)–(16)).

Continuous target ratio. Substitute $V_{ITS} = \tau r_D D/(r_U - g)$ into (3):

$$r_{WACC} - g = (r_U - g) - \frac{\tau r_D D}{V_L}$$

$$r_{WACC} = r_U - \tau r_D \frac{D}{D+E} \tag{4}$$

and with (12) of note 01, $r_U = \frac{E}{V} r_E + \frac{D}{V} r_D$, substituted into (4):

$$r_{WACC} = \left(\frac{E}{V} r_E + \frac{D}{V} r_D\right) - \tau r_D \frac{D}{V} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D (1-\tau) \tag{5}$$

— formula (1) recovered: **(1) is exact under a continuously rebalanced target ratio.**

Annual target ratio. Substitute (16) of note 01 into (3):

$$r_{WACC} = r_U - \tau r_D \frac{D}{D+E} \cdot \frac{1+r_U}{1+r_D} \tag{6}$$

(slide 18 prints $1 + r_g$ again; it is $1 + r_D$, as derived in note 01 §8).

Constant debt level. Substitute $V_{ITS} = \tau D$ into (3):

$$r_{WACC} = g + (r_U - g)\left(1 - \tau \frac{D}{D+E}\right) \tag{7}$$

and with $g = 0$ — the only case in which a constant debt is also a constant D/V —

$$r_{WACC} = \left(1 - \tau \frac{D}{D+E}\right) r_U \tag{8}$$

which is slide 18's first line (Ross's 9.589% in L03-01-tax-shield-mm-taxes §5; Lista R3 Ex. 13, Ross 16.19, printed p. 578). **With $g > 0$ and a constant debt, D/V drifts down, so no constant WACC exists** — use APV.

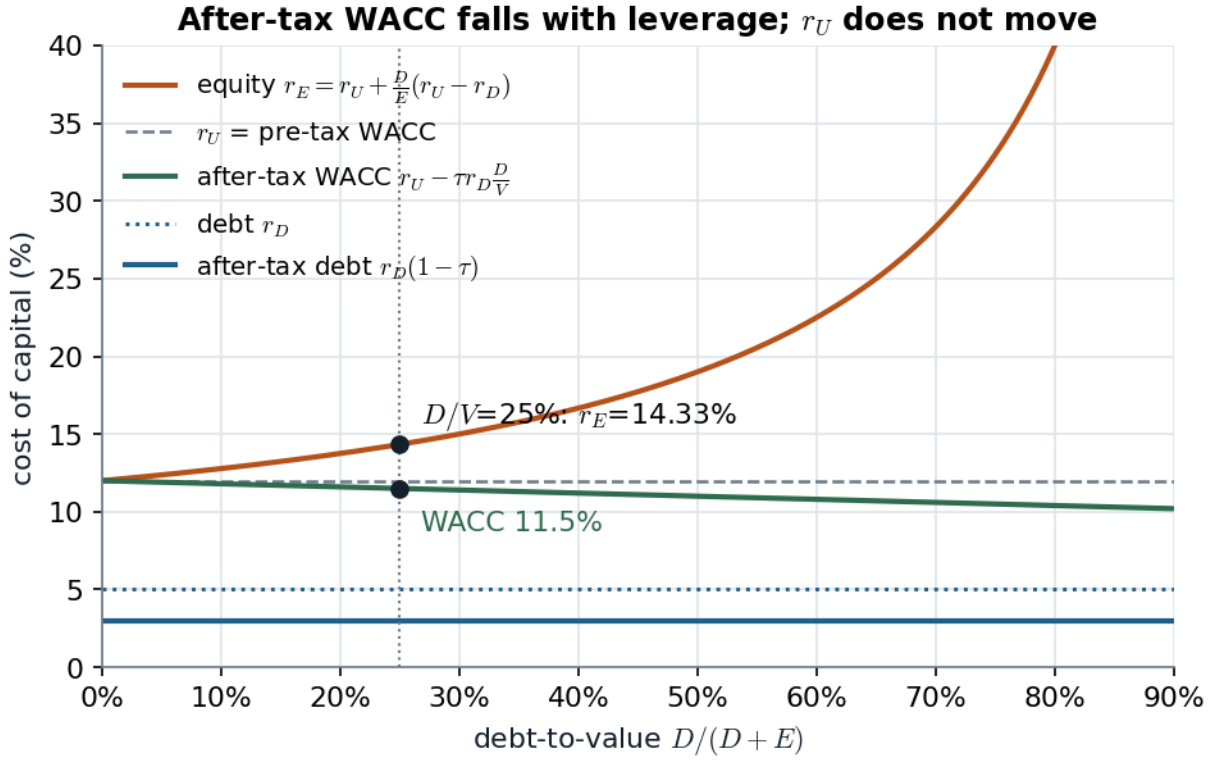


Figure — the slide-17 picture (Berk Figure 16.7, p. 478) redrawn with the slide-28 project's numbers

and safe debt: r_E from (13) of note 01, after-tax WACC from (4). At $D/V = 25\%$: $r_E = 14.33\%$, $r_{WACC} = 11.5\%$. Source: slides 12, 16–18, 28–30.

r_U — the pre-tax WACC — is flat: the assets' risk does not depend on financing. The after-tax WACC falls linearly in D/V with slope $-\tau r_D = -2$ points per 100% of leverage; Berk's version curves upward at high leverage only because there r_D rises with default risk.

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3 · C — the two methods, step by step (slides 19–24)

Step	**APV** (slides 20–21)	**WACC** (slide 22)
1	unlevered after-tax FCF	unlevered after-tax FCF
2	unlevered β_U from comparables	unlevered β_U from comparables
3	V_U: discount FCF at r_U	**relever** β_U at the target D/E $\rightarrow r_E$
4	add V_{ITS}: PV of tax savings on current and future **incremental** interest	r_D from similar projects' debt
5	—	after-tax WACC (1); $V_L = \sum_t FCF_t/(1+r_{WACC})^t$ (slide 23)

APV details (slide 21): a temporary increase in debt → value the shield **as an annuity, not a perpetuity**; use the **appropriate** r_{ITS}; use the **marginal** tax rate (the firm's rate *including* the project). Higgins (p. 322) writes APV as

$$APV = NPV_{\text{all-equity}} + PV_{\text{interest tax shields}} + PV_{\text{any other side effects}}$$

— *"valuation by parts"*, needed "when the firm's capital structure is changing over time, or when the project's debt capacity differs from that implicit in the WACC" (pp. 321–322).

Slide 24, the map:

Method	Cash flows	Discount rate
APV	unlevered FCF	unlevered (asset) cost of capital r_U
	interest tax shield	tax-shield cost of capital r_{ITS}
WACC	unlevered FCF	after-tax WACC r_{WACC}

If leverage changes a lot over time, r_{WACC} must be computed period by period (slide 23).

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4 · C — equivalence when D/E is constant (slides 25–27)

WACC method with growth g:

$$V_L = \frac{FCF_1}{r_{WACC} - g} \tag{9}$$

Substitute (1), written with $V_L = D + E$:

$$V_L = \frac{FCF_1}{\frac{E}{V_L} r_E + \frac{D}{V_L} r_D (1-\tau) - g} \tag{10}$$

Split $r_D(1-\tau) = r_D - r_D\tau$ and use (12) of note 01, $\frac{E}{V_L} r_E + \frac{D}{V_L} r_D = r_U$ (constant D/E):

$$V_L = \frac{FCF_1}{r_U - \frac{D}{V_L} r_D \tau - g} \tag{11}$$

Cross-multiply:

$$V_L(r_U - g) - D r_D \tau = FCF_1 \tag{12}$$

Move $D r_D \tau$ right and divide by $r_U - g$:

$$\boxed{V_L = \underbrace{\frac{FCF_1}{r_U - g}}_{=V_U} + \underbrace{\frac{D r_D \tau}{r_U - g}}_{=V_{ITS}}} \tag{13}$$

Equivalence (slide 27). (13) is APV with the continuous-target shield (15) of note 01. The one assumption used is (12) of note 01 — D/E **constant, hence** $r_{ITS} = r_U$. Drop it and step (10)→(11) fails.

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5 · Worked — the 60,000 project, both ways (slides 28–31)

Data. Investment 60,000 today; unlevered FCF 7,500 in year 1, growing 4% (7,500 → 7,500 × 1.04 = 7,800 → 7,800 × 1.04 = 8,112 ...); $\tau = 40\%$; target $D/(D+E) = 25\%$; $r_E = 14.33\%$, $r_D = 5\%$ pre-tax.

WACC method.

$$r_{WACC} = 0.75 \times 14.333\% + 0.25 \times 5\% \times (1 - 0.40) = 10.75\% + 0.75\% = 11.5\%$$

(with the slide's rounded 14.33%: $10.7475\% + 0.75\% = 11.4975\%$, printed 11.5%; the exact r_E is $12\% + \frac{1}{3}(12\% - 5\%) = 14.333\ldots\%$ from (13) of note 01.)

$$V_L = \frac{7{,}500}{0.115 - 0.04} = \frac{7{,}500}{0.075} = 100{,}000, \qquad NPV = 100{,}000 - 60{,}000 = 40{,}000$$

APV method. $D = 0.25 \times 100{,}000 = 25{,}000$ (slide 30). Unlevered cost of capital, from (12) of note 01:

$$r_U = 0.75 \times 14.333\% + 0.25 \times 5\% = 10.75\% + 1.25\% = 12\%$$

$$V_U = \frac{7{,}500}{0.12 - 0.04} = \frac{7{,}500}{0.08} = 93{,}750$$

Debt grows at 4% with the firm, so the shield does too, discounted at r_U (slide 31):

$$V_{ITS} = \frac{25{,}000 \times 0.05 \times 0.40}{0.12 - 0.04} = \frac{500}{0.08} = 6{,}250$$

$$V_L = 93{,}750 + 6{,}250 = 100{,}000, \qquad NPV = 40{,}000 \checkmark$$

6 · Worked — APV's flexibility: a fixed 60,000 loan repaid in year 5 (slide 32)

New policy: borrow 60,000 at $t = 0$, pay 5% interest, repay at $t = 5$. V_U is unchanged at 93,750.

Year	0	1	2	3	4	5
Net borrowing	60,000					−60,000
Interest (5%)		−3,000	−3,000	−3,000	−3,000	−3,000
ITS (40%)		1,200	1,200	1,200	1,200	1,200

The schedule is fixed in currency, so the shield is as risky as the debt: $r_{ITS} = r_D = 5\%$, an annuity (slide 21):

$$\frac{1 - 1.05^{-5}}{0.05} = \frac{1 - 0.783526}{0.05} = 4.329477$$

$$V_{ITS} = 1{,}200 \times 4.329477 = 5{,}195.37 \approx 5{,}195$$

$$V_L = 93{,}750 + 5{,}195 = 98{,}945, \qquad NPV = 38{,}945$$

Why no WACC here. D/V_L starts at 60,000/98,945 = 61% and drops to zero after year 5: there is no single r_{WACC} to plug into (9). Berk's Example 16.4 (p. 476) is the same calculation: 2 billion of ten-year 6% bonds, $\tau = 35\%$, shield $0.35 \times 120 = 42$ million a year, $PV = 42 \times \frac{1}{0.06}(1 - \frac{1}{1.06^{10}}) = 42 \times 7.360087 = 309$ million.

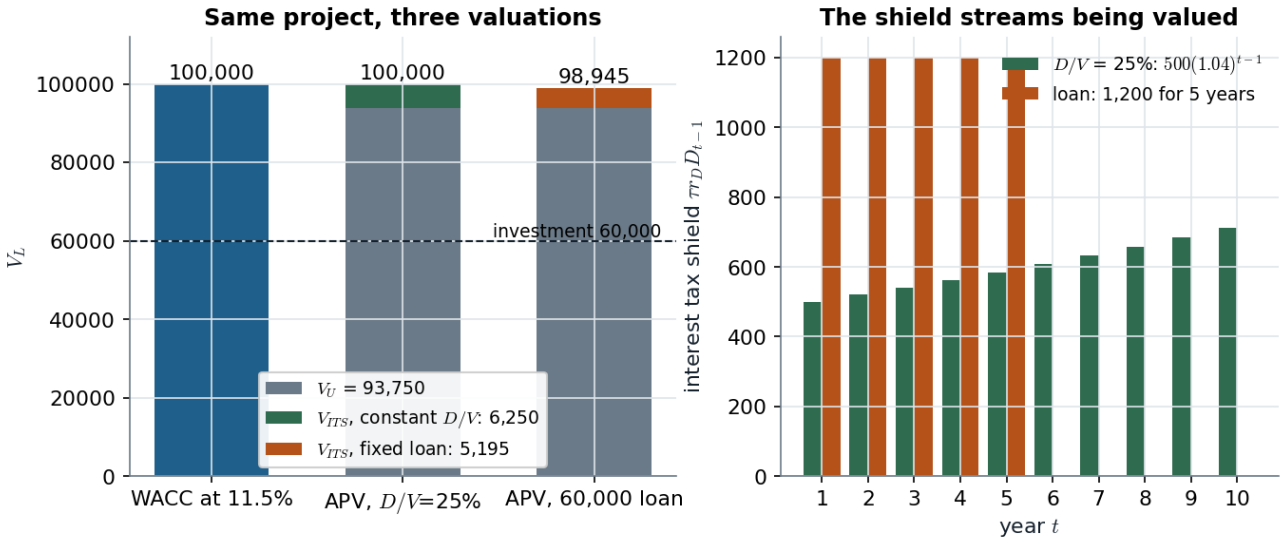


Figure — the slide-28 project valued three ways; right panel: the two shield streams being discounted, $500(1.04)^{t-1}$ at r_U and 1,200 for five years at r_D. Source: slides 28–32; eqs. (9), (13).

The loan's shields are larger early (1,200 against 500) yet worth less in total because they stop after five years; the target-ratio shields are small, risky and perpetual. **What the shield is worth depends on the debt policy, which is why APV asks for it explicitly.**

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7 · C — Appendix: the WACC formula derived (slides 54–61)

Assumptions (slide 54): $V_L = D + E = V_U + V_{ITS}$; the firm keeps a constant leverage ratio (continuously adjusted); r_U is not time-varying. Then the expected return on the levered firm's assets equals that of its claims,

$$r_L = \frac{D}{D+E} r_D + \frac{E}{D+E} r_E \tag{14}$$

and, under the constant-ratio assumption, $r_L = r_U$ in every period (note 01 (12)).

Cash flow and one-period value (slide 55). Investors receive the unlevered FCF plus the shield:

$$C_{L,t+1} = FCF_{t+1} + r_D D_t \tau \tag{15}$$

$$V_{L,t} = \frac{FCF_{t+1} + r_D D_t \tau + V_{L,t+1}}{1 + r_L} \tag{16}$$

Iterate (slides 56–57). (16) one period later: $V_{L,t+1} = \frac{FCF_{t+2} + r_D D_{t+1} \tau + V_{L,t+2}}{1+r_L}$. Substituting it into (16):

$$V_{L,t} = \frac{C_{L,t+1}}{1+r_L} + \frac{C_{L,t+2}}{(1+r_L)^2} + \frac{V_{L,t+2}}{(1+r_L)^2}$$

After N substitutions:

$$V_{L,t} = \sum_{i=1}^{N} \frac{C_{L,t+i}}{(1+r_L)^i} + \frac{V_{L,t+N}}{(1+r_L)^N}$$

and with $V_{L,t+N}/(1+r_L)^N \to 0$ as $N \to \infty$ (slide 57):

$$V_{L,t} = \sum_{i=1}^{\infty} \frac{C_{L,t+i}}{(1+r_L)^i} \tag{17}$$

— the going-concern formula; unusable in practice because it needs every period's shield (slide 58).

Fold the shield into the rate (slides 59–60). Multiply (16) by $1 + r_L$ and move the shield left:

$$V_{L,t}(1+r_L) - r_D D_t \tau = FCF_{t+1} + V_{L,t+1}$$

Write $r_D D_t \tau = V_{L,t}\, r_D \frac{D_t}{V_{L,t}} \tau$ and factor $V_{L,t}$:

$$V_{L,t}\Big(1 + r_L - r_D \frac{D_t}{V_{L,t}} \tau\Big) = FCF_{t+1} + V_{L,t+1}$$

$$V_{L,t} = \frac{FCF_{t+1} + V_{L,t+1}}{1 + r_L - r_D \frac{D_t}{V_{L,t}} \tau} \tag{18}$$

Expand the denominator with (14) and $V_{L,t} = D_t + E_t$:

$$1 + r_L - r_D \frac{D_t}{V_{L,t}} \tau = 1 + \frac{D_t}{D_t + E_t} r_D + \frac{E_t}{D_t + E_t} r_E - r_D \frac{D_t}{D_t + E_t} \tau$$

$$= 1 + \frac{D_t}{D_t + E_t} r_D (1 - \tau) + \frac{E_t}{D_t + E_t} r_E = 1 + r_{WACC} \tag{19}$$

Result (slide 61). Substituting (19) into (18), $V_{L,t} = \frac{FCF_{t+1} + V_{L,t+1}}{1 + r_{WACC}}$, and the same iteration as (17) gives

$$\boxed{V_{L,t} = \sum_{i=1}^{\infty} \frac{FCF_{t+i}}{(1 + r_{WACC})^i}} \tag{20}$$

Read (19) carefully. The step $r_L = r_U$ was not even needed for (19) — it is an identity in $D_t/V_{L,t}$. What makes (20) useful is that r_{WACC} **is the same number every period**, which requires a constant $D_t/V_{L,t}$ (and constant r_E, r_D, which the constant ratio and constant r_U deliver). Slide 61's summand prints $FCF_{L,t+i}$; it is the unlevered FCF_{t+i}.

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8 · C — which r_D goes into the shield? (slides 62–63)

The expected return on debt is defined by

$$D_t = \frac{E[C_{t+1} + D_{t+1}]}{1 + r_D} \iff r_D = \frac{E[C_{t+1} + D_{t+1}] - D_t}{D_t} \tag{21}$$

Safe bond (slide 62): issued at par, face 1,000, coupon 5%: $E[C_{t+1}] = 50$, $E[D_{t+1}] = 1{,}000$, so $r_D = \frac{1050 - 1000}{1000} = 5\%$ — coupon rate = expected return.

Risky bond (slide 63): $E[C_{t+1} + D_{t+1}] = 700 < 1{,}050$ and $D_t = 525$:

$$r_D = \frac{700 - 525}{525} = 33.33\%$$

Is the expected interest deduction $525 \times 33.33\% = 175$? **No.** The deduction is the **interest actually paid** — the coupon — so the rate that sets the shield is

$$\bar{r}_D = \frac{E[C_{t+1}]}{D_t}, \qquad \text{after-tax cost of debt} = r_D - \tau\, \bar{r}_D \tag{22}$$

Illustration (own assumption: the 50 coupon is paid, so $E[C_{t+1}] = 50$ — the slide does not split the 700; $\tau = 35\%$ as on slide 2): $\bar{r}_D = 50/525 = 9.52\%$, after-tax cost $33.33\% - 0.35 \times 9.52\% = 33.33\% - 3.33\% = 30.00\%$, against the naive $33.33\% \times (1 - 0.35) = 21.67\%$. **Using $r_D(1-\tau)$ with a distressed bond overstates the shield by a factor of $175/50 = 3.5$.** The expected return includes compensation for expected default losses (here principal); those are not deductible interest. Compare L02 §2A.10: the yield is not r_D either.

Links

- Before: L03-01-tax-shield-mm-taxes (the three V_{ITS} closed forms, r_{ITS}).
- After: L03-04-static-trade-off §6 — APV with a distress term, $V_L = V_U + V_{ITS} - PV$(bankruptcy costs).
- Topic 2A: unlevering comparables and the pre-tax WACC, L02 MM II and WACC.
- Rules: rules/03.
- Practice: PS2 Ex. 2(a)–(e) (PS2, p. 2: r_E before and after, new WACC, shield under a continuous target, interest as 20% of FCF, fixed perpetual debt); Lista R3 Ex. 4 (Higgins 8.23, p. 333), Ex. 6 (Berk 16.11, p. 492), Ex. 9 (Grinblatt–Titman 13.15–13.16, p. 494), Ex. 13 (Ross 16.19, p. 578).
- Code: L03-08-python §1–§2.

2B.5–2B.6 · The 100%-debt puzzle and the costs of financial distress

> **Claim.** MM with taxes says every unit of debt adds τ to value, so the optimum is all debt — and no industry is near it (slide 34: debt is 12–44% of value). The missing piece is **not** default itself: handing the firm to creditors is just a change of owner and costs nothing in MM. **It is the deadweight costs that default brings** — direct (lawyers, managers' time, fire sales) and indirect (lost customers, suppliers, workers, and conflicts that distort investment) — and these can offset the tax advantage.

Prev: L03-02-after-tax-wacc-apv · Next: L03-04-static-trade-off · Map: L03-00-map

Notation. V_U, V_L, τ, D as in L03-01-tax-shield-mm-taxes; K the promised debt repayment; X the cash flow at maturity; C the deadweight cost of default.

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1 · C — the puzzle in one line (slide 33)

An unlevered firm issues debt D, fixed in perpetuity, to repurchase equity, with $\tau = 35\%$. By MM I with taxes (L03-01-tax-shield-mm-taxes (8), (5)):

$$V_L = V_U + 0.35D \tag{1}$$

Divide by V_U:

$$\frac{V_L}{V_U} = 1 + 0.35\frac{D}{V_U} \tag{2}$$

D/V_U	V_L/V_U	value gain
20%	$1 + 0.35 \times 0.20 = 1.07$	**+7%**
50%	$1 + 0.35 \times 0.50 = 1.175$	**+17.5%**
100%	1.35	+35%

(2) is increasing in D without bound: *"it seems that the optimal capital structure is attained with 100% debt. Obviously this is not true"* (slide 33). Ross reaches the same "conclusão ilógica" (p. 553); Berk asks *"why don't firms use debt financing almost exclusively?"* (p. 478); Amaro (§2.1.4, p. 47): *"In reality, firms are never fully financed with debt."*

2 · U — what firms actually do (slides 34–35)

Slide 34 — Jagannathan et al. (2017), Table 5, WACC by Fama–French industry, 2016 (selected columns, read off the slide):

Industry	β_S	T_C (%)	$E[R_S]$ (%)	$E[R_D]$ (%)	$D/(D+S)$	WACC (%)
Durable	1.42	30.71	11.74	4.61	**0.44**	8.02
High-tech	0.98	20.60	8.87	4.22	**0.12**	8.21
Health	0.74	18.35	7.28	4.28	0.15	6.66
Utilities	0.53	32.32	5.93	4.61	**0.41**	4.69
Telecom	0.68	35.69	6.90	4.55	0.27	5.86

Recompute one row with (1) of note 02 (Utilities): $0.59 \times 5.93\% + 0.41 \times 4.61\% \times (1 - 0.3232) = 3.499\% + 1.279\% = 4.78\%$ against the printed 4.69% — the table averages firm-level WACCs (market-cap weighted), so the industry averages of the inputs do not reproduce it exactly. **The reading that matters: even the most levered industry sits at 44% debt, the tech industry at 12%.**

Ross Table 16.7 (p. 567, Ibbotson 2008, debt/total capital): pharmaceuticals 6.76%, computers 5.31%, steel 28.57%, airlines 56.30%, cable TV 61.84%, electric utilities 49.40%.

Slide 35 — Graham, Leary and Roberts (2015). US corporate leverage 1920–2010: book debt + preferred over capital roughly 20–50%, market leverage roughly 10–35% (read off the figure), rising sharply after the 1960s. **A century of data, and never near 100%.**

> **Two facts for any theory to fit (Berk p. 482):** firms choose debt "too low to fully exploit the interest tax shield", and leverage differs systematically **across industries** — low in tech and pharma, high in utilities and cable.

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3 · C — default is not the cost (slide 36)

"A bankruptcy process transfers ownership to debtholders. This transfer of control is not a violation of MM per se. The violation comes from bankruptcy costs."

Why the transfer is free. In the MM world of Topic 2A, debt is a claim $\min(X, K)$ and equity $\max(X - K, 0)$ (L02 risky debt and equity as a call). They always sum to X: in default creditors simply receive all of X. Nothing is lost, so V does not depend on K. Ross (p. 556): *"Em uma situação de mercados perfeitos, não há custos associados a essa transferência de propriedade, e os credores nada perderiam."*

With a deadweight cost C in default, the claims sum to $X - C$ in default states and to X otherwise. The expected loss $C \times \Pr(X < K)$ grows with K: **that** is what can "countervail the tax advantage of debt financing" (slide 36) — made exact in L03-04-static-trade-off (7).

4 · C — direct costs (slide 37)

Ex post, conditional on actual liquidation or reorganization:

- legal and administrative costs;
- time and resources of managers and creditors dealing with the bankruptcy;
- fire sales of assets, due to asset specificity and illiquid assets.

Magnitudes. Berk (p. 479): direct costs "approximately 3% to 4% of the pre-bankruptcy market value of total assets", higher for smaller and more complex firms; Enron's fees exceeded 750 million. Ross (p. 557): Enron spent more than 1 billion on lawyers, accountants and consultants; WorldCom 600 million. Amaro (p. 48), citing Warner (1977) on 11 railroads: about **5.3 cents per dollar**, with a strong inverse size effect — 1.7% for the largest. **Small relative to a 35% tax rate** — which is why the indirect costs carry the argument.

What a bankruptcy cost is not (Higgins p. 210): ACE's apartment complex lost value through local overbuilding; that loss happens whatever the financing, so it is *not* a bankruptcy cost. The cost is only what bankruptcy itself adds — high when assets have low **resale value** (Moletek, a genetic-engineering firm whose assets are a research team and growth options), low for redeployable, tangible assets (pp. 210–211).

5 · C — indirect costs (slides 38–40)

Ex ante: incurred **before** any filing, as the probability of distress rises.

Channel	Mechanism	Evidence on the slides
Customers	demand falls where the product needs the firm to survive — airlines (frequent-flier programs), autos (warranties, parts, service)	Hortaçsu et al. (2013): a rise in a manufacturer's CDS lowers its used-car auction prices, more so for longer-lived cars
Suppliers and banks	credit from suppliers and banks dries up (credit-market imperfections)	—
Workers	risk-averse workers want stable jobs and avoid unemployment scarring	Brown & Matsa (2016): fewer applications per opening in distress; Baghai et al. (2021): firms lose the most skilled workers near bankruptcy, and talent-dependent firms choose more conservative capital structures; Van Doornik et al. (2022): more generous UI shifts labour supply to riskier firms and shrinks the wage premium they must pay
Equity–debt conflicts	poor investment decisions: **debt overhang** (underinvestment), **asset substitution / risk-shifting** (excessive risk), **cashing out** (withdrawing cash ahead of senior claimants)	—

Berk (p. 480) adds fire sales and puts the total at **10–20% of firm value** (Andrade and Kaplan's highly levered firms); Higgins (p. 211) stresses that indirect costs feed on one another — cut-backs in R&D and marketing, lost sales, nervous suppliers, competitors who "smell blood".

5.1 · U — the two investment distortions, in numbers (own illustration)

The slides name these conflicts; their formal treatment is the agency-cost topic (syllabus item 4, Tirole Ch. 3). The numbers below are an own illustration, risk-neutral investors, $r_f = 0$, no taxes.

Risk-shifting. Debt promises $K = 100$. The firm can keep a safe asset paying 90, or switch to a gamble paying 160 or 0 with probability 1/2 each (mean 80, **worse by 10**).

	Safe (90)	Gamble (160 / 0)
Equity $E[\max(X-100,0)]$	$\max(90-100,0)=0$	$\frac{1}{2}(60)+\frac{1}{2}(0)=30$
Debt $E[\min(X,100)]$	90	$\frac{1}{2}(100)+\frac{1}{2}(0)=50$
Total	90	80

Shareholders gain 30 by destroying 10 of value — creditors lose 40. Equity is a call option on the assets, and a call gains from volatility (Higgins p. 212's "go-for-broke" problem).

Debt overhang. Same $K = 100$; assets in place pay 60 or 120 (prob. 1/2). A new project costs 20 today and pays 30 for sure (NPV +10), financed by the shareholders.

	Without project	With project (assets pay 90 / 150)
Equity	$\frac{1}{2}(0) + \frac{1}{2}(20) = 10$	$\frac{1}{2}(0) + \frac{1}{2}(50) = 25$
Debt	$\frac{1}{2}(60) + \frac{1}{2}(100) = 80$	$\frac{1}{2}(90) + \frac{1}{2}(100) = 95$

Shareholders put in 20 to raise their claim by $25 - 10 = 15$: they **reject** a project with NPV +10, because 15 of its 30 goes to creditors (Higgins p. 212, *"debt overhang"*).

> **These are costs of debt even if no court is ever involved**: creditors anticipate the distortion and price it into the debt, so shareholders bear it ex ante (Ross p. 557–558: *"o efeito líquido é a perda de valor, porque a empresa optou por usar dívida"*).

6 · C — the logic that the trade-off model formalizes

Force	Grows with leverage because	Direction
Tax shield	more interest deducted	**more debt**
Expected distress cost	probability of default × cost if it happens (Higgins p. 209: "the probability bankruptcy will occur times the costs incurred when it does")	**less debt**

Higgins's checklist (p. 213) is the whole model in words: (1) the firm's **ability to use** the extra shields, (2) the **increased probability** of distress, (3) the **magnitude** of distress costs. In L03-04-static-trade-off these become $1 - H(K)$, $h(K)$ and C in one first-order condition.

7 · G — Miller's objection and DeAngelo–Masulis (Amaro §2.2, pp. 48, 54–55; named, not examinable)

Miller (1977) argued that bankruptcy costs are **too small** to balance a 48-cent tax gain per dollar (the 20% figure people cited applied to individuals; Warner found 5.3%), and proposed personal taxes instead (L03-01-tax-shield-mm-taxes §10). DeAngelo and Masulis (1980) restore an interior optimum from the **supply** side: with uncertain earnings, more borrowing raises the chance that **some shields go unused**, so the effective τ_c falls with debt; add leverage-related deadweight costs $d(B)$ and the supply curve of corporate bonds slopes down (pp. 55–56). The slides' model has the first effect built in: no tax is paid in default, so an extra unit of K saves tax only with probability $1 - H(K)$.

Links

- Before: L03-01-tax-shield-mm-taxes, L03-02-after-tax-wacc-apv. Next: L03-04-static-trade-off.
- Topic 2A: L02 MM I, risky debt, equity as a call.

- Forward: debt overhang and risk-shifting are the agency frictions of syllabus item 4 (lectures-index, "Bridges forward").
- Bankruptcy procedure (named only): Ross §16.10, pp. 567–568 (liquidation vs. reorganization, absolute priority rule).
- Practice: PS2 Ex. 3(c) (PS2, p. 2: a direct cost of 7% of the realized cash flow — who bears it); Lista R3 Ex. 8 (Berk 16.22, p. 494), Ex. 10 (Grinblatt–Titman 16.2, pp. 591–592); quiz block `dist` of quiz Topic 2B.

2B.7 · The static trade-off: $V(K)$, the first-order condition, and K^*

Claim. In one period with risk-neutral investors, a firm that promises K to creditors is worth $V(K) = V_U + \frac{1}{1+r_f}[\tau\, E[\min(X, K)] - C\, H(K)]$: **the tax saved on everything creditors are paid, minus the expected default cost.** Raising K saves τ in solvent states only and raises the default probability, so the optimum equates $\tau[1 - H(K^*)] = C\, h(K^*)$. K^* **rises with τ and falls with C; with $C = 0$ the firm wants all the debt it can issue, with $\tau = 0$ any safe debt will do — MM again.**

Prev: L03-03-debt-puzzle-bankruptcy-costs · Next: L03-05-bank-capital-dividend-irrelevance · Map: L03-00-map

Notation (slides 41–42).

Symbol	Meaning	Range
$t = 0, 1$	debt issued at 0, repaid at 1	
K	**total** repayment promised at $t = 1$ (principal + interest)	≥ 0
X	random cash flow at $t = 1$, density h on $[\underline{X}, \bar{X}]$	$0 \leq \underline{X} < \bar{X}$
$H(a)$	c.d.f., $\Pr(X \leq a)$; $H(K)$ is the **default probability**	$[0, 1]$
τ	tax rate, paid only if solvent ($X \geq K$); the **full** K is deductible	$[0, 1)$
C	fixed deadweight cost of bankruptcy ($X < K$), with $C < \underline{X}$	≥ 0
r_f	required expected return of every (risk-neutral) investor	> -1
V_U, $E(K)$, $D(K)$, $V(K)$	unlevered value; market values of equity, debt, and the firm	
$S(K)$	$E[\min(X, K)] = \int_{\underline{X}}^{K} x h(x) dx + K[1 - H(K)]$	
K^*	value-maximizing K	$[\underline{X}, \bar{X}]$

Which assumption does what. The full deductibility of K is "for simplicity" (slide 41) — it makes the shield of a solvent firm exactly τK. No tax in default means the shield there is τX — the whole tax bill. $C < \underline{X}$ keeps creditors' payoff $X - C$ positive, so limited liability never binds. Risk neutrality makes $1/(1 + r_f)$ the only discount factor.

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1 · C — the unlevered firm and safe debt (slide 42)

The default probability (slide 41):

$$H(a) = \Pr(X \leq a) = \int_{\underline{X}}^{a} h(x)\, dx \tag{1}$$

Unlevered. The firm pays τX in every state; investors get $(1-\tau)X$:

$$V_U = \frac{1}{1+r_f}\int_{\underline{X}}^{\bar{X}} (1-\tau)X\,h(X)\,dX = \frac{(1-\tau)E[X]}{1+r_f} \tag{2}$$

Safe debt, $K < \underline{X}$. Default never happens. Equity gets $(1-\tau)(X-K)$, debt gets K:

$$V_L(K) = \frac{1}{1+r_f}\int_{\underline{X}}^{\bar{X}} \big[(1-\tau)(X-K)+K\big]h(X)\,dX$$

Integrate each term ($\int h = 1$):

$$V_L(K) = \frac{(1-\tau)E[X-K]}{1+r_f} + \frac{K}{1+r_f}$$

Write $(1-\tau)E[X-K]+K = (1-\tau)E[X]+\tau K$:

$$V_L(K) = V_U + \frac{\tau K}{1+r_f} \qquad (K < \underline{X}) \tag{3}$$

— **one-period MM I with taxes**: the shield τK is safe, discounted at r_f (L03-01-tax-shield-mm-taxes (5) is the perpetual version).

2 · C — risky debt: equity, debt, firm (slides 43–44)

Equity (slide 43) — paid only when solvent, after tax:

$$E(K) = \frac{1}{1+r_f}\int_{K}^{\bar{X}} (1-\tau)(X-K)\,h(X)\,dX \tag{4}$$

Debt — in default creditors take X and bear C; when solvent they get K:

$$D(K) = \frac{1}{1+r_f}\int_{\underline{X}}^{K} (X-C)\,h(X)\,dX + \frac{1}{1+r_f}\int_{K}^{\bar{X}} K\,h(X)\,dX$$

and $\int_K^{\bar{X}} h = 1 - H(K)$:

$$D(K) = \frac{1}{1+r_f}\int_{\underline{X}}^{K} (X-C)\,h(X)\,dX + \frac{K[1-H(K)]}{1+r_f} \tag{5}$$

(Slide 43 labels the first line "the market value of debt" twice; (4) is equity.)

Adding (4) and (5) (slide 44), every step. Put both over $1+r_f$:

$$(1+r_f)V(K) = \int_{K}^{\bar{X}} \big[(1-\tau)(X-K)+K\big]h\,dX + \int_{\underline{X}}^{K} (X-C)h\,dX$$

Rewrite each integrand around the unlevered payoff $(1-\tau)X$: $(1-\tau)(X-K)+K = (1-\tau)X+\tau K$ and $X - C = (1-\tau)X + \tau X - C$. Then

$$(1+r_f)V(K) = \int_{\underline{X}}^{\bar{X}} (1-\tau)X\,h\,dX + \int_{K}^{\bar{X}} \tau K\,h\,dX + \int_{\underline{X}}^{K} (\tau X - C)\,h\,dX$$

The first integral is $(1+r_f)V_U$ by (2); the second is $\tau K[1-H(K)]$; the third splits into $\tau\int_{\underline{X}}^{K} Xh\,dX - C\,H(K)$ by (1). Dividing by $1+r_f$:

$$\boxed{V(K) = V_U + \frac{1}{1+r_f}\Big[\tau\Big(\int_{\underline{X}}^{K} x\,h(x)\,dx + K[1-H(K)]\Big) - C\,H(K)\Big]} \tag{6}$$

Read (6). The round bracket is $S(K) = E[\min(X,K)]$ — what the firm pays out to creditors before the cost C:

$$V(K) = V_U + \underbrace{\frac{\tau\,S(K)}{1+r_f}}_{V_{ITS}} - \underbrace{\frac{C\,H(K)}{1+r_f}}_{PV(\text{bankruptcy costs})} \tag{7}$$

which is slide 45's $V_L = V_U + V_{ITS} - PV(\text{bankruptcy costs})$, Berk's Eq. (16.10) (p. 481), and Ross's Figure 16.6 (p. 559) in a formula.

A second reading. From (5), $\int_{\underline{X}}^{K} x\,h\,dx + K[1-H(K)] = (1+r_f)D(K) + C\,H(K)$. Substitute into (6):

$$V(K) = V_U + \tau D(K) - \frac{(1-\tau)\,C\,H(K)}{1+r_f} \tag{8}$$

With $C = 0$:

$$V(K) = V_U + \tau D(K) \tag{9}$$

— **Amaro's Proposition 2.1.3** (p. 45): with no deadweight cost, risky debt still adds exactly τ times its market value. **Default alone costs nothing** (L03-03-debt-puzzle-bankruptcy-costs §3).

3 · C — the marginal condition (slides 44–45)

Differentiate each piece of (6) in K. By Leibniz's rule,

$$\frac{d}{dK}\int_{\underline{X}}^{K} x\,h(x)\,dx = K\,h(K)$$

$$\frac{d}{dK}\Big(K[1-H(K)]\Big) = [1-H(K)] - K\,h(K)$$

Their sum is $1-H(K)$ — the $K\,h(K)$ terms cancel: the creditors' gain in the marginal state exactly offsets the loss of the marginal solvent state. And $\frac{d}{dK}C\,H(K) = C\,h(K)$. So

$$(1+r_f)\,V'(K) = \underbrace{\tau\,[1-H(K)]}_{\text{marginal tax benefit}} - \underbrace{C\,h(K)}_{\text{marginal expected bankruptcy cost}} \tag{10}$$

Slide 44's "two effects". An extra unit of promised repayment (i) saves τ, but only in **solvent** states — probability $1-H(K)$; in default states all tax is already avoided; (ii) pushes probability mass $h(K)$ into default, each unit costing C.

First-order condition (slide 45). *"The optimal debt level K^* is such that the marginal benefit of debt equals the marginal expected bankruptcy costs"*:

$$\boxed{\tau\,[1-H(K^*)] = C\,h(K^*)} \tag{11}$$

Second-order condition. Differentiate (10) again:

$$(1+r_f)\,V''(K) = -\tau\,h(K) - C\,h'(K) < 0 \tag{12}$$

at an interior maximum.

The safe region. For $K < \underline{X}$, $H = h = 0$ and (10) gives $V'(K) = \tau/(1+r_f) > 0$: **the firm never stops below** $\underline{X}$ — it always takes all the safe debt it can, so $K^* \geq \underline{X}$.

4 · C — comparative statics: K^* rises with τ, falls with C (slide 45)

Write (11) as $F(K^*;\tau,C) = 0$ with $F \equiv \tau[1-H(K)] - C\,h(K)$. The partial derivatives:

$$F_K = -\tau h(K) - Ch'(K), \qquad F_\tau = 1-H(K), \qquad F_C = -h(K)$$

By the implicit function theorem $dK^*/dx = -F_x/F_K$, and $F_K < 0$ by (12):

$$\frac{dK^*}{d\tau} = \frac{1-H(K^*)}{\tau h(K^*) + Ch'(K^*)} > 0 \tag{13}$$

$$\frac{dK^*}{dC} = \frac{-h(K^*)}{\tau h(K^*) + Ch'(K^*)} < 0 \tag{14}$$

> **Slide 45: "K^* increases with the tax benefits τ; K^* decreases with bankruptcy costs C."** (13)–(14) are those two sentences, signed by the second-order condition alone — they hold for any density h that makes the optimum interior.

The two landmarks.

Case	(10) becomes	Optimum
$C = 0$	$\tau[1 - H(K)] \geq 0$, zero only at $K = \bar{X}$	$K^* = \bar{X}$: promise the top cash flow, **shield the whole tax bill**; $V = V_U + \tau E[X]/(1 + r_f) = E[X]/(1 + r_f)$ — the 100%-debt answer of L03-03-debt-puzzle-bankruptcy-costs §1
$\tau = 0$	$-C\,h(K) \leq 0$	**any** $K \leq \underline{X}$ gives V_U (MM indifference over safe debt); risky debt only loses $C\,H(K)/(1 + r_f)$

5 · Worked — uniform cash flow (the companion's calibration)

The slides leave h general and give no numbers. **Calibration used here and in the companion (declared, not the slides'):** $X \sim U[\underline{X}, \bar{X}] = U[50, 150]$, $\tau = 35\%$ (the rate of slides 2 and 33), $C = 20 < \underline{X}$, $r_f = 5\%$ (the debt rate of slide 32).

Uniform pieces. $h = \frac{1}{\bar{X} - \underline{X}}$, $H(K) = \frac{K - \underline{X}}{\bar{X} - \underline{X}}$, and for $\underline{X} \leq K \leq \bar{X}$:

$$S(K) = \frac{K^2 - \underline{X}^2}{2(\bar{X} - \underline{X})} + \frac{K(\bar{X} - K)}{\bar{X} - \underline{X}} \tag{15}$$

Optimum. Substitute h and H into (11):

$$\tau\,\frac{\bar{X} - K^*}{\bar{X} - \underline{X}} = \frac{C}{\bar{X} - \underline{X}}$$

Multiply by $\bar{X} - \underline{X}$ and solve:

$$K^* = \bar{X} - \frac{C}{\tau} \quad \text{if } C < \tau(\bar{X} - \underline{X}); \qquad K^* = \underline{X} \text{ otherwise} \tag{16}$$

($h' = 0$, so (12) is $-\tau h < 0$: the interior root is a maximum.) **Uniform h makes (13)–(14) explicit: $dK^*/d\tau = C/\tau^2 > 0$, $dK^*/dC = -1/\tau < 0$.**

The numbers, step by step.

$$V_U = \frac{(1 - 0.35) \times 100}{1.05} = \frac{65}{1.05} = 61.9048$$

$$K^* = 150 - \frac{20}{0.35} = 150 - 57.1429 = 92.8571 \qquad (20 < 0.35 \times 100 = 35 : \text{ interior})$$

$$H(K^*) = \frac{92.8571 - 50}{100} = 0.428571$$

$$S(K^*) = \frac{92.8571^2 - 50^2}{200} + \frac{92.8571 \times 57.1429}{100} = \frac{8622.45 - 2500}{200} + 53.0612 = 30.6122 + 53.0612 = 83.6735$$

$$V_{ITS} = \frac{0.35 \times 83.6735}{1.05} = \frac{29.2857}{1.05} = 27.8912$$

$$PV(\text{bankruptcy costs}) = \frac{20 \times 0.428571}{1.05} = \frac{8.5714}{1.05} = 8.1633$$

$$V(K^*) = 61.9048 + 27.8912 - 8.1633 = 81.6327$$

The claims. By (5): $\int_{50}^{K^*}(x-20)\frac{dx}{100} = \frac{(8622.45-2500)/2 - 20\times 42.8571}{100} = \frac{3061.22 - 857.14}{100} = 22.0408$, so

$$D(K^*) = \frac{22.0408 + 92.8571 \times 0.571429}{1.05} = \frac{22.0408 + 53.0612}{1.05} = \frac{75.1020}{1.05} = 71.5258$$

By (4): $E(K^*) = \frac{0.65}{1.05} \cdot \frac{(150-92.8571)^2}{2\times 100} = \frac{0.65\times 16.3265}{1.05} = 10.1069$.

Check: $71.5258 + 10.1069 = 81.6327 = V(K^*)$ ✓; and (8): $61.9048 + 0.35 \times 71.5258 - \frac{0.65\times 8.5714}{1.05} = 61.9048 + 25.0340 - 5.3061 = 81.6327$ ✓.

Debt policy	V	vs. V_U
none ($K = 0$)	61.9048	—
largest safe debt ($K = \underline{X} = 50$)	$61.9048 + \frac{0.35\times 50}{1.05} =$ $61.9048 + 16.6667 = 78.5714$	+16.67
optimum $K^* = 92.86$	**81.6327**	**+19.73**
default certain ($K \geq 150$)	$61.9048 + \frac{0.35\times 100 - 20}{1.05} =$ $61.9048 + 14.2857 = 76.1905$	+14.29

At K^*: default probability 42.9%, $D/V = 71.53/81.63 = 87.6\%$, promised yield $K^*/D - 1 = 92.857/71.526 - 1 = 29.8\%$ against an **expected** return of $r_f = 5\%$ — the gap between yield and expected return of L02 §2A.10.

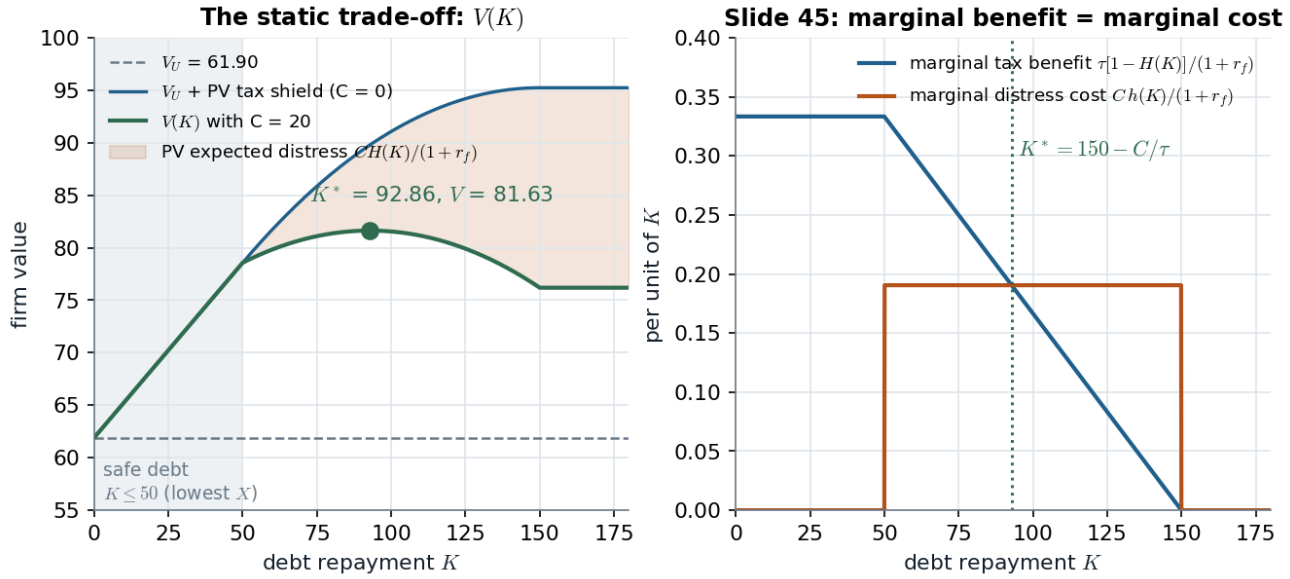


Figure — the trade-off model with uniform h on $[50, 150]$, $\tau = 35\%$, $C = 20$, $r_f = 5\%$: $V(K)$ from (6), the tax-shield-only curve ($C = 0$, eq. (9)), and the marginal condition (10)–(11). $K^* = 92.86$, $V^* = 81.63$. Source: slides 42–45; calibration declared in §5.

The shaded wedge is $C\,H(K)/(1 + r_f)$: zero while debt is safe, then growing linearly with the default probability. On the right, the marginal tax benefit is flat at $\tau/(1 + r_f) = 0.333$ over safe debt and declines as solvent states run out; the distress cost jumps to $C/(100 \times 1.05) = 0.190$ once K enters the support. Where they cross is K^*.

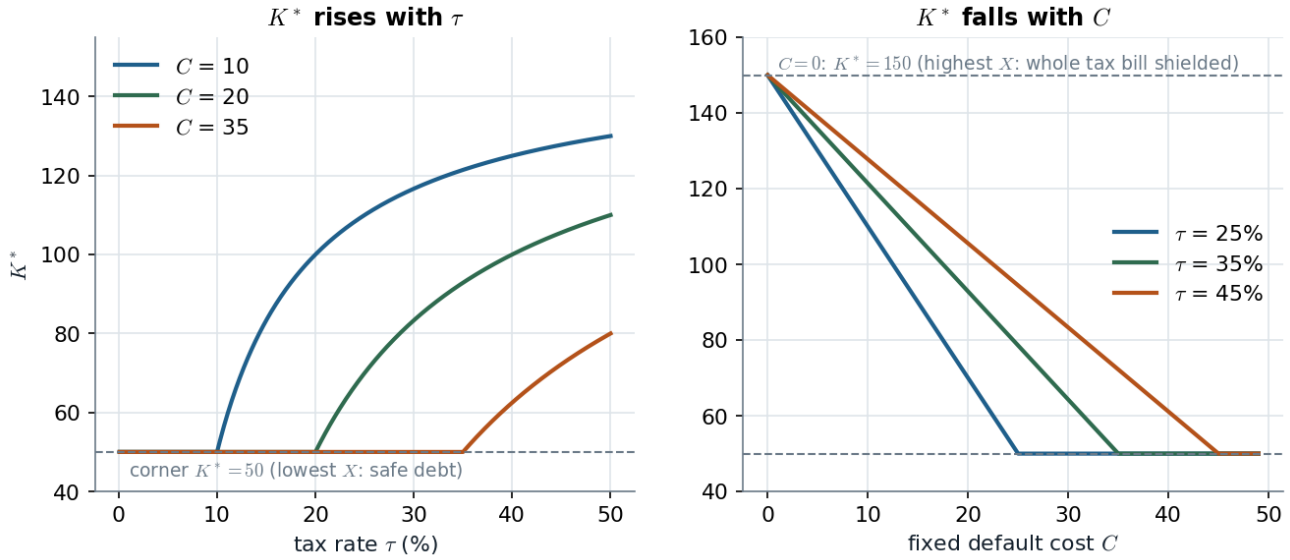


Figure — slide 45's comparative statics under (16): K^* rises with τ (left) and falls with C (right); the kinks are the switch to the corner $K^* = \underline{X}$ at $C = \tau(\bar{X} - \underline{X})$. Source: eqs. (13)–(16).

Both landmarks of §4 are visible: at $C = 0$ every curve on the right starts at $\bar{X} = 150$, and as $\tau \to 0$ every curve on the left falls to the safe-debt corner.

Interactive companion. The Trade-off Dial (`companions/companion-static-trade-off.html`) — move τ, C, the support $[\underline{X}, \bar{X}]$ and r_f, watch $V(K)$ and the two marginal curves move and read K^*, $V(K^*)$, the shield, the expected distress cost and the default

probability; set $C = 0$ or $\tau = 0$ to land on the two landmarks.

6 · C — from the model to the APV and to practice (slide 45; Ross pp. 560–562)

"We can adjust the APV method to consider PV(bankruptcy costs)" (slide 45): (7) is APV with a third term — Higgins's "any other side effects" (L03-02-after-tax-wacc-apv §3).

What the model says to a CFO — Ross's recommendations (pp. 560–562) and Higgins's three factors (p. 213), each one an object of (10):

Practical advice	Where it sits in (10)
The shield matters only for firms that **pay tax**: tax-loss carry-forwards, other shields (depreciation) and, in Brazil, non-*lucro real* regimes lower the gain (Ross p. 560, 562)	the effective τ
Higher tax rate → more borrowing (Ross p. 562)	(13)
Riskier firms borrow less: "quanto maior a volatilidade do Lajir, menos uma empresa deveria tomar emprestado" (Ross p. 562)	H, h — see the caveat below
Intangible assets, growth options, talent → costly distress → less debt (Ross p. 562; Higgins pp. 210–211)	C, (14)

> **Caveat (an exact property of the model, not in the slides).** The slides sign K^* in τ and C only. Volatility is **not** signed in general: with uniform h, $K^* = \bar{X} - C/\tau$ depends on the top of the support, so a mean-preserving spread **raises** K^* in the interior regime and **lowers** it in the corner regime $K^* = \underline{X}$. What does fall with volatility in the uniform case, in both regimes, is the value gain from debt, $V(K^*) - V_U$ (mean 100, $\tau = 35\%$, $C = 20$: 30.00 at width 20, 19.73 at width 100, 18.17 at width 140 — computed in L03-08-python §3). Pitfall 7 in L03-07-pitfalls.

Why "static". Ross (p. 558): the theory takes the firm's assets and operations as fixed and varies only the debt–equity mix — one period, one decision, no refinancing, no information problems.

7 · G — the tax-shield underutilization link (Amaro §2.2.5, pp. 54–55)

DeAngelo and Masulis's force is already in (10): with $C = 0$ the marginal benefit $\tau[1 - H(K)]$ still **declines** with debt because shields are wasted in default states. On its own it never turns negative (so $K^* = \bar{X}$); it is the deadweight cost $C\,h(K)$ that produces an interior optimum — Amaro's $d(B)$ (p. 55).

Links

- Before: L03-03-debt-puzzle-bankruptcy-costs (what C stands for), L03-01-tax-shield-mm-taxes (the shield).
- Topic 2A: MM I with risky debt and put–call parity, L02 MM I; the two-state entrepreneur, L02 entrepreneur example.
- Rules: rules/03 (taxes and default costs).
- Practice: Lista R3 Ex. 8 (Berk 16.22, p. 494), Ex. 10 (Grinblatt–Titman 16.2, pp. 591–592: the two-state version of this model); PS2 Ex. 3(c) (PS2, p. 2); quiz block `dist` of quiz Topic 2B.

- Code: L03-08-python §3 (general h, numerical K^*, a triangular density).
- Companion: The Trade-off Dial (`companions/companion-static-trade-off.html`).

2B.8–2B.9 · Bank capital, and why payout policy is irrelevant too

> **Claim.** The lecture's tools settle a live policy debate: of the three standard objections to higher bank equity requirements, **two are MM fallacies** (a lower ROE and "expensive" equity leave the WACC unchanged) and **one is a private cost that is a public transfer** (the lost tax shield). The same zero-NPV logic then extends to payout: **with the investment plan fixed, a higher dividend must be financed by issuing claims, and value is $V_t = E[\sum_i \delta^i (X_{t+i} - I_{t+i})]$ whatever the payout.**

Prev: L03-04-static-trade-off · Next: L03-06-formulas · Map: L03-00-map

Notation. §1: r_U, r_E, r_D, τ, D/V, E/V as in L03-01-tax-shield-mm-taxes; ROE here is the expected return on book-equal-to-market equity. §2 (slides 49–51): X_t random cash flow of period t; D_t **per-share dividend** (not debt!); I_t investment; n_t shares outstanding at the end of t; P_t ex-dividend share price at the end of t; δ the discount factor; $V_t = n_t P_t$.

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1 · C — the bank capital debate (slides 46–47)

After the 2007–08 crisis regulators proposed higher capital (equity) requirements. Banks argued this would hurt credit and growth. Admati, DeMarzo, Hellwig and Pfleiderer (2013): **"bank equity is not socially expensive"** — larger capital requirements, less leverage, using the arguments of this lecture. The slide asks: true or false?

Illustration (own, built on the slide-28 rates): a bank's assets earn $r_U = 12\%$, its debt is safe at $r_D = 5\%$; the requirement moves E/V from 25% to 50%.

Statement 1 — *"Increased equity requirements will hurt bank shareholders since it would lower the banks' return on equity (ROE)."* Without taxes, by MM II (L03-01-tax-shield-mm-taxes (13)):

$$r_E\big|_{E/V=25\%} = 12\% + \frac{0.75}{0.25}(12\% - 5\%) = 12\% + 3 \times 7\% = 33\%$$

$$r_E\big|_{E/V=50\%} = 12\% + \frac{0.50}{0.50}(12\% - 5\%) = 19\%$$

ROE falls — **but so does the risk it compensates**: with $\beta_D = 0$, $\beta_E = \frac{V}{E}\beta_U$ halves from $4\beta_U$ to $2\beta_U$. 33% was the fair return for a riskier claim; 19% is the fair return for a safer one. Shareholders' wealth is unchanged (MM I). **False** — the ROE–leverage link of Topic 1 (L01 ROE and leverage) is a statement about **risk**, not value.

Statement 2 — *"… increase the funding costs for banks because they must use more equity, which has a higher required return."* The WACC with no taxes:

$$0.25 \times 33\% + 0.75 \times 5\% = 8.25\% + 3.75\% = 12\%, \qquad 0.50 \times 19\% + 0.50 \times 5\% = 9.5\% + 2.5\% = 12\%$$

False — the required return on equity is not a fixed price: it **falls** as leverage falls, exactly offsetting the shift in weights (L02 MM II).

Statement 3 — *"… because they reduce the ability of banks to benefit from the tax shield associated with interest payments on debt."* With $\tau = 40\%$ and a continuous target, by (4) of L03-02-after-tax-wacc-apv:

$r_{WACC}\big|_{D/V=75\%} = 12\% - 0.40 \times 5\% \times 0.75 = 10.5\%, \qquad r_{WACC}\big|_{D/V=50\%} = 12\% - 0.40 \times 5\% \times 0.50 = 11.0\%$

True for the bank, but not a social cost: the 0.5 point is the tax shield $\tau r_D D$, a transfer from taxpayers. Less leverage moves it back; the economy's resources are the same. Slide 47's counterpoint completes the case: leverage carries a **negative externality** — it "raises the probability of a bank failure, weakening the financial system and imposing losses on the broader economy" — a cost of debt the bank does not bear, so the private trade-off (L03-04-static-trade-off (11)) chooses **too much** leverage from society's view.

> **The pattern.** A claim that "equity is expensive" is either an MM fallacy (statements 1–2) or a claim about a subsidy (statement 3). Neither is a social cost of equity.

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2 · C — MM dividend irrelevance (slides 48–51)

Slide 48. The MM logic applies to **all** purely financial transactions, since each has zero NPV: other instruments (convertibles, preferred stock), the choice between long- and short-term debt, risk management — and payout. ***MM dividend irrelevance proposition:*** *a firm's total market value is independent of its dividend policy.* (Berk p. 508: "holding fixed the investment policy of a firm, the firm's choice of dividend policy is irrelevant and does not affect the initial share price"; Amaro **Proposition 4.1.1 (MM3)**, p. 100.)

Setup (slide 49). Investment policy fixed ⇒ a change in payout must be met by a change in financing. All-equity firm, risk-neutral investors; at $t = 1, 2, ...$ it generates X_t, pays a per-share dividend D_t, invests I_t, has n_t shares.

Pricing (slide 50). The end-of-t price is the discounted value of next period's dividend and price:

$$P_t = \delta E[D_{t+1} + P_{t+1}] \tag{1}$$

Sources = uses (slide 50). New shareholders of period t buy at P_t and do not receive D_t, so the dividend to the **old** shares plus investment is paid by the cash flow plus the issue:

$$\underbrace{n_{t-1} D_t + I_t}_{\text{uses of funds}} = \underbrace{X_t + P_t(n_t - n_{t-1})}_{\text{sources (earned or raised)}} \tag{2}$$

Total value (slide 51). Multiply (1) by n_t:

$$V_t = n_t P_t = \delta E[n_t D_{t+1} + n_t P_{t+1}] \tag{3}$$

Write (2) for period $t+1$ and solve for the dividend bill:

$$n_t D_{t+1} = X_{t+1} - I_{t+1} + P_{t+1}(n_{t+1} - n_t) \tag{4}$$

Substitute (4) into (3):

$$V_t = \delta E[X_{t+1} - I_{t+1} + P_{t+1}(n_{t+1} - n_t) + n_t P_{t+1}]$$

The $n_t P_{t+1}$ terms cancel, leaving $P_{t+1} n_{t+1} = V_{t+1}$:

$$V_t = \delta E[X_{t+1} - I_{t+1} + V_{t+1}] \tag{5}$$

Solve recursively. (5) at $t+1$ inside (5) at t: $V_t = \delta E[X_{t+1} - I_{t+1}] + \delta^2 E[X_{t+2} - I_{t+2}] + \delta^2 E[V_{t+2}]$; after N steps the remainder is $\delta^N E[V_{t+N}] \to 0$ (Amaro p. 100: "Assuming that V is always finite and r is positive"), so

$$\boxed{V_t = E\Big[\sum_{i=1}^{+\infty} \delta^i \left(X_{t+i} - I_{t+i}\right)\Big]} \tag{6}$$

(slide 51 prints the summand as $X_{t+1} - I_{t+1}$; it is $X_{t+i} - I_{t+i}$, as in Amaro's (4.5), p. 100). **No D appears in (6).** The value of the all-equity firm does not depend on the payout policy **as long as the investment plan I_t is fixed**; the result extends to debt issuance and to risk aversion (slide 51).

The substitution is the economics (Amaro p. 100): "an increase in dividends is exactly compensated by the issue of new equity, so that both effects cancel out". Amaro's (4.7) (p. 101) shows the dividend view is the same number: $V_t = \sum_\tau \frac{D_{t+\tau}}{(1+r)^\tau} - \sum_\tau \frac{m_{t+\tau} p_{t+\tau}}{(1+r)^\tau}$ — dividends minus the PV of the equity raised to pay them.

3 · Worked — two textbook firms, every policy

Ross's Wharton Corporation (§17.2, p. 588). All-equity, 100 shares, cash flow 10,000 in each of the next two years (then dissolved), required return 10%.

Policy A — dividend = cash flow: 100 per share each year.

$$P_0 = \frac{100}{1.10} + \frac{100}{1.10^2} = 90.909 + 82.645 = 173.55$$

Policy B — 110 per share at date 1: 11,000 paid against 10,000 earned, so 1,000 of new shares are sold at date 1. The new holders require $1{,}000 \times 1.10 = 1{,}100$ at date 2, leaving $10{,}000 - 1{,}100 = 8{,}900$ for the old shares, 89 per share:

$$P_0 = \frac{110}{1.10} + \frac{89}{1.10^2} = 100.000 + 73.554 = 173.55$$

Homemade dividends (p. 589): an investor who wanted 100 and 100 reinvests 10 of the 110 at 10% and has $89 + 11 = 100$ at date 2; one who wanted 110 and 89 sells 10 of stock at date 1 and gives up $10 \times 1.1 = 11$.

Berk's Genron (§17.2, pp. 503–507). 20 million of excess cash, no debt, 10 million shares, FCF 48 million a year forever, $r_U = 12\%$: enterprise value $48/0.12 = 400$, total value 420.

Policy	Today	Each later year	Share price
1 — pay the cash as a dividend	2.00	$48/10 = 4.80$	cum $2 + 4.80/0.12 = 2 + 40 = 42$; ex 40
2 — repurchase $20/42 = 0.476$ m shares	0	$48/9.524 = 5.04$	$5.04/0.12 = 42$
3 — issue $28/42 = 0.67$ m shares, pay 48 now	$48/10.67 = 4.50$	4.50	$4.50 + 4.50/0.12 = 4.50 + 37.50 = 42$

42 in every row (Berk Table 17.1, p. 507). *"Although dividends do determine share prices, a firm's choice of dividend policy does not"* (p. 508). Berk's two COMMON MISTAKE boxes are the pitfalls: a repurchase does not raise the price "because supply falls" (assets fall too, p. 506), and the **bird-in-the-hand** view that higher current dividends raise the price is wrong in perfect markets (p. 508).

> **The one sentence.** A payout is a financial transaction: it moves value between the firm's cash and the shareholders' pockets, and between old and new shareholders at a fair price; only the investment plan $\{I_t\}$ — and frictions such as taxes (Berk §17.3) — can change V.

Links

- Before: L03-04-static-trade-off (the private trade-off that the externality distorts); L03-01-tax-shield-mm-taxes (MM II with and without taxes).
- Topic 1 and 2A: L01 ROE and leverage; L02 MM II and WACC.
- Rules: rules/03 (MM3, Amaro p. 100).
- Practice: PS2 Ex. 1(f) (PS2, p. 1: a debt-financed dividend in a no-tax world); Lista R3 Ex. 2 (Berk 17.7, p. 528); quiz block `pol` of quiz Topic 2B.
- Code: L03-08-python §4 (the recursion (5) checked on random payout policies).

L03 · Formulas — every definition, result and number

Prev: L03-05-bank-capital-dividend-irrelevance · Next: L03-07-pitfalls · Map: L03-00-map

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0 · The exam specification, mapped

The slides carry no learning-goal list; the specification is the lecture's block map (lectures-index, Topic 2B), item by item.

Block	What you must be able to do	Note
2B.1	compute the ITS from two income statements; say why unlevered FCF ignores it; name the friction (differential treatment)	L03-01-tax-shield-mm-taxes §1–§3
2B.2	derive $V_L = V_U + \tau D$ for perpetual safe debt; write the market-value balance sheet and both r_E formulas; derive the three V_{ITS} closed forms; price a debt-financed repurchase	L03-01-tax-shield-mm-taxes §4, §7–§9
2B.3	write the after-tax WACC; derive the three WACC formulas	L03-02-after-tax-wacc-apv §1–§2
2B.4	run APV and WACC; prove their equivalence under constant D/E; value a fixed loan by APV	L03-02-after-tax-wacc-apv §3–§6
2B.5	state the 100%-debt puzzle and the evidence against it	L03-03-debt-puzzle-bankruptcy-costs §1–§2
2B.6	explain why default is not an MM violation; classify direct vs. indirect costs; name the three equity–debt conflicts	L03-03-debt-puzzle-bankruptcy-costs §3–§5
2B.7	derive $E(K)$, $D(K)$, $V(K)$; the FOC; sign $dK^*/d\tau$, dK^*/dC; solve the uniform case	L03-04-static-trade-off
2B.8	judge the three bank-capital statements	L03-05-bank-capital-dividend-irrelevance §1
2B.9	derive $V_t = E\sum \delta^i(X_{t+i} - I_{t+i})$ from sources = uses	L03-05-bank-capital-dividend-irrelevance §2–§3
2B.10	derive the WACC from $V_{L,t} = (FCF_{t+1} + \tau r_D D_t + V_{L,t+1})/(1 + r_L)$; use $\bar{r}_D$, not r_D, for a risky bond's shield	L03-02-after-tax-wacc-apv §7–§8

1 · Definitions and results

#	Statement	Assumption that drives it	Derived in
ITS	$ITS = \tau r_D D$ (Berk 16.4)	interest deductible	01 (1)
Perpetual safe debt	per-period payout $(1-\tau)X + \tau r_f D$; $V_{ITS} = \tau D$	constant, safe, permanent D	01 (2)–(5)
MM I with taxes	$V_L = V_U + V_{ITS}$ (Ross 16.3; Berk 16.5; Amaro Prop. 2.1.3, $V_l^\tau = V_u^\tau + \tau B$)	no other friction	01 (8)
Balance-sheet identity	$\frac{D}{V} r_D + \frac{E}{V} r_E = \frac{V_U}{V_L} r_U + \frac{V_{ITS}}{V_L} r_{ITS}$; same for betas	$V_L = D + E = V_U + V_{ITS}$	01 (9)–(10)
MM II, constant debt	$r_E = r_U + \frac{D}{E}(1-\tau)(r_U - r_D)$ (Ross 16.4); Hamada $\beta_E = \beta_U[1 + (1-\tau)D/E]$	$r_{ITS} = r_D$	01 (11)
MM II, target D/E	$r_E = r_U + \frac{D}{E}(r_U - r_D)$; $\beta_U = \frac{E}{V}\beta_E + \frac{D}{V}\beta_D$	$r_{ITS} = r_U$ ("usual assumption")	01 (12)–(13)
V_{ITS}, constant debt	τD	$r_{ITS} = r_D$	01 (14)
V_{ITS}, continuous target	$\frac{\tau r_D D}{r_U - g}$	$r_{ITS} = r_U$	01 (15)
V_{ITS}, annual target	$\frac{\tau r_D D}{r_U - g} \cdot \frac{1+r_U}{1+r_D}$ (slide prints $1 + r_g$)	shield known one period ahead	01 (16)
After-tax WACC	$r_{WACC} = \frac{E}{V} r_E + \frac{D}{V} r_D (1-\tau)$ (Berk 16.8)	continuous target D/E	02 (1), (5)
General WACC	$r_{WACC} - g = (r_U - g)(1 - V_{ITS}/V_L)$	growing perpetuity	02 (3)
WACC, continuous / annual	$r_U - \tau r_D \frac{D}{V}$; $r_U - \tau r_D \frac{D}{V} \frac{1+r_U}{1+r_D}$		02 (4), (6)
WACC, constant debt	$(1 - \tau \frac{D}{V}) r_U$	$g = 0$ **only**	02 (7)–(8)
APV = WACC	$\frac{FCF_1}{r_{WACC} - g} = \frac{FCF_1}{r_U - g} + \frac{D r_D \tau}{r_U - g}$	constant D/E	02 (9)–(13)
APV (Higgins p. 322)	$NPV_{\text{all-equity}} + PV_{ITS} + PV_{\text{side effects}}$	—	02 §3
Appendix	$V_{L,t} = \frac{FCF_{t+1} + r_D D_t \tau + V_{L,t+1}}{1+r_L} \Rightarrow V_{L,t} = \sum_i \frac{FCF_{t+i}}{(1+r_{WACC})^i}$	constant ratio, constant r_U	02 (14)–(20)
Shield of risky debt	$\bar{r}_D = E[C_{t+1}]/D_t$; after-tax cost $r_D - \tau \bar{r}_D$	deduction = coupon paid	02 (21)–(22)
Puzzle	$V_L/V_U = 1 + \tau D/V_U$	MM + taxes only	03 (2)
Trade-off: V_U	$\frac{(1-\tau)E[X]}{1+r_f}$	risk neutrality	04 (2)
Safe debt	$V_L(K) = V_U + \frac{\tau K}{1+r_f}$, $K < \underline{X}$		04 (3)

#	Statement	Assumption that drives it	Derived in
Equity, debt	$E(K) = \frac{1}{1+r_f}\int_K^{\bar{X}}(1-\tau)(X-K)h$; $D(K) = \frac{1}{1+r_f}[\int_{\underline{X}}^{K}(X-C)h + K(1-H(K))]$	tax only if solvent; $C < \underline{X}$	04 (4)–(5)
Firm value	$V(K) = V_U + \frac{1}{1+r_f}[\tau(\int_{\underline{X}}^{K} xh + K(1-H)) - CH(K)]$		04 (6)
APV with distress	$V_L = V_U + V_{ITS} - PV$(bankruptcy costs) (Berk 16.10)		04 (7)
Second reading	$V = V_U + \tau D(K) - \frac{(1-\tau)CH(K)}{1+r_f}$; $C = 0 \Rightarrow V_U + \tau D(K)$		04 (8)–(9)
Marginal value	$(1+r_f)V'(K) = \tau[1-H(K)] - Ch(K)$	Leibniz	04 (10)
FOC	$\tau[1-H(K^*)] = Ch(K^*)$		04 (11)
SOC; statics	$-\tau h - Ch' < 0$; $dK^*/d\tau > 0$, $dK^*/dC < 0$	SOC	04 (12)–(14)
Uniform h	$K^* = \bar{X} - C/\tau$ if $C < \tau(\bar{X} - \underline{X})$, else $\underline{X}$		04 (16)
MM3	$V_t = E[\sum_{i\geq 1}\delta^i(X_{t+i} - I_{t+i})]$ (Amaro Prop. 4.1.1, (4.5))	investment plan fixed	05 (1)–(6)

2 · Every number in the lecture

Case	Data	Result	Note
Two firms (slide 2, Berk Table 16.2)	EBIT 1850, interest 350, τ 35%	taxes 525 vs 647.5; NI 975 vs 1202.5; **ITS 122.5**; FCF 1502.5 both	01 §1–§2
Ross firms U, L (pp. 551–553)	EBIT 1000, D 1000 at 8%, τ 30%, R_U 10%	ITS 24; V_{ITS} 300; V_U 7,000; V_L 7,300; R_E 10.22%; WACC 9.59%	01 §5
Higgins Timid/Bold (p. 225)	EBIT 400, 800 debt at 10%, τ 40%	80% vs 96% for 200 of equity; gap 32 = 0.4 × 80	01 §6
Repurchase (slides 14–15)	V_U 600, 1000 shares, D 500, τ 40%	V_L 800, E 300, **P = 0.8**, N = 625, gain 200	01 §9
Slide-13 forms at slide-28 data	D 25,000, r_D 5%, τ 40%, r_U 12%, g 4%	6,250 (continuous); 6,666.67 (annual); 10,000 (fixed)	01 §8

Case	Data	Result	Note
Project (slides 28–31)	60,000; FCF 7,500, g 4%; τ 40%; D/V 25%; r_E 14.33%; r_D 5%	**WACC 11.5%**, V_L 100,000, NPV 40,000; r_U 12%, V_U 93,750, V_{ITS} 6,250	02 §5
Fixed loan (slide 32)	60,000 at 5%, repaid year 5	ITS 1,200 $\times$ 5; V_{ITS} 5,195; V_L **98,945**	02 §6
Berk Example 16.4 (p. 476)	2 bn, 6%, 10 years, τ 35%	42 m/yr; PV 309 m	02 §6
Risky bond (slides 62–63)	face 1,000, coupon 5%; $E[\cdot]$ 700, D 525	r_D 33.33%; with coupon 50: $\bar{r}_D$ 9.52%	02 §8
Puzzle (slide 33)	τ 35%	+7% at D = 20% V_U; +17.5% at 50%	03 §1
Jagannathan et al. (slide 34)	10 industries, 2016	$D/(D+S)$ 0.12 (high-tech) – 0.44 (durable)	03 §2
Trade-off, uniform (own calibration)	$U[50, 150]$, τ 35%, C 20, r_f 5%	V_U 61.90; K^* **92.86**; H 0.429; V_{ITS} 27.89; PV(bc) 8.16; V^* **81.63**; D 71.53; E 10.11	04 §5
Wharton (Ross p. 588)	10,000 $\times$ 2 years, 100 shares, 10%	173.55 under 100/100 and 110/89	05 §3
Genron (Berk pp. 503–507)	FCF 48 m, 12%, cash 20 m, 10 m shares	42 under all three policies	05 §3

L03 · Pitfalls and an original True/False self-test

Prev: L03-06-formulas · Next: L03-08-python · Map: L03-00-map

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Part I — the confusions this lecture produces

1 · Counting the shield twice

Unlevered FCF already excludes the shield (L03-01-tax-shield-mm-taxes §2). **Either** discount it at the after-tax WACC **or** discount it at r_U and add V_{ITS} — never both. *Minimal counterexample:* slide-28 project at 11.5% plus 6,250 gives 106,250, not 100,000.

2 · Using the levered FCF in the WACC

Discounting $FCF + \tau r_D D$ at the after-tax WACC is the same double count from the other side: the appendix's (16) uses r_L, (18) uses r_{WACC}, and the shield moves from the numerator to the denominator — it is not in both (L03-02-after-tax-wacc-apv §7).

3 · Mixing the two MM II formulas

$r_E = r_U + \frac{D}{E}(1-\tau)(r_U - r_D)$ is for a **fixed debt level**; $r_E = r_U + \frac{D}{E}(r_U - r_D)$ for a **target ratio**. Unlevering comparables with Hamada and relevering into a WACC built on a target ratio mixes two incompatible policies. *Counterexample:* at $D/E = 1/3$, $r_U = 12\%$, $r_D = 5\%$, $\tau = 40\%$ the two give 13.40% and 14.33%.

4 · Applying $(1 - \tau D/V)r_U$ to a growing firm

That WACC is exact only for constant debt **and** $g = 0$ (L03-02-after-tax-wacc-apv (7)–(8)). With $g > 0$ a constant debt level means a falling D/V: use APV.

5 · "Default is what makes debt costly"

The transfer of control is free in MM (L03-03-debt-puzzle-bankruptcy-costs §3); with $C = 0$ the trade-off model gives $V = V_U + \tau D(K)$, rising all the way to $K = \bar{X}$ (L03-04-static-trade-off (9)). **The cost is the deadweight C, not the default.**

6 · Reading $\tau[1 - H(K)]$ as "the tax shield"

It is the **marginal** benefit of one more unit of promised repayment. The **level** of the shield is $\tau E[\min(X, K)]$, which includes τX in default states. The marginal benefit declines because in default states the extra unit shields nothing — all tax is already avoided.

7 · "More volatile firms always choose less debt"

Slide 45 signs K^* in τ and C only. In the uniform case $K^* = \bar{X} - C/\tau$ **rises** with a mean-preserving spread in the interior regime and falls in the corner regime; what falls with volatility there is the **gain** from debt (L03-04-static-trade-off §6). Ross's advice is about debt capacity in practice, not a theorem of this model.

8 · Using r_D for a distressed bond's shield

The deduction follows the coupon paid, $\bar{r}_D = E[C_{t+1}]/D_t$; the expected return r_D includes compensation for default losses that are not interest (slides 62–63: $525 \times 33.33\% = 175 \neq$ the interest).

9 · "A lower ROE hurts shareholders" / "equity is expensive"

ROE and r_E fall with leverage **because risk falls**; the WACC is unchanged without taxes (L03-05-bank-capital-dividend-irrelevance §1). Only the tax subsidy makes equity privately costlier, and that is a transfer.

10 · D_t in slides 49–51 is a dividend per share

Not debt. And the dividend-irrelevance result needs the **investment plan fixed**: a dividend financed by cutting a positive-NPV project does lower value.

Part II — True/False self-test (24 items, original)

Score as in `rules/00_estilo_avaliacao.md`: $\max\{0,\ \text{right} - \text{wrong}/2\} \times$ item value.

1. *With corporate taxes, a levered firm pays less tax than an identical unlevered firm, but its net income is lower.* — **T.** Interest reduces taxable income (taxes 525 vs 647.5 on slide 2), and net income falls by interest minus the tax saved ($1202.5 - 975 = 227.5 = 350 \times 0.65$). Total payments to investors rise by $350 \times 0.35 = 122.5$; equity alone does not.

2. *The interest tax shield arises because corporate taxes reduce every claim on the firm proportionally.* — **F.** A proportional tax on all claims would leave MM intact. The shield exists because interest is deducted and dividends and retained earnings are not (slide 5): the friction is the **differential** treatment.

3. *Unlevered free cash flow already includes the interest tax shield, so it must be discounted at r_U to value the levered firm.* — **F.** FCF uses $EBIT(1-\tau)$, as if there were no interest (slide 4); discounting it at r_U gives V_U, and the shield must be added (APV) or put into the rate (after-tax WACC).

4. *With perpetual safe debt, $V_{ITS} = \tau D$ regardless of the interest rate.* — **T.** The shield $\tau r_f D$ is a riskless perpetuity at r_f; the rate cancels: $\tau r_f D / r_f = \tau D$.

5. *Under MM I with taxes, the gain from a debt-financed repurchase is shared between the shareholders who sell and the creditors.* — **F.** Creditors pay a fair price for a fair claim. The whole V_{ITS} goes to the original shareholders — sellers included, since they sell at the post-announcement price (0.8 on slide 15; gain $200 = V_{ITS}$).

6. *In the slide-14 repurchase the share price rises when the shares are bought back.* — **F.** It rises at the announcement, when the market values the shield; the buyback then happens at 0.8 and leaves the price unchanged ($300/375 = 0.8$).

7. *Investors can replicate the tax advantage of corporate debt by borrowing on personal account.* — **F.** Personal interest is not deductible at the corporate rate (Higgins Table 6A.1: 80% on the homemade replica vs. 96% on Bold). That is why investors pay a premium for levered firms (slide 8).

8. *If a firm keeps a constant debt-to-equity ratio, the tax shields are as risky as the firm's assets.* — **T.** Debt moves with asset value, so do the shields: $\beta_{ITS} = \beta_U$, $r_{ITS} = r_U$ (slide 12).

9. *Under a target D/E, the cost of equity is $r_U + \frac{D}{E}(1-\tau)(r_U - r_D)$.* — **F.** That is the constant-debt formula. With $r_{ITS} = r_U$ the balance sheet gives $\frac{D}{V}r_D + \frac{E}{V}r_E = r_U$, hence $r_E = r_U + \frac{D}{E}(r_U - r_D)$ — the no-tax formula.

10. *For a growing firm, a shield worth $\tau r_D D/(r_U - g)$ is always worth less than τD.* — **F.** It is less when $r_U - g > r_D$ and more when growth is high enough that $r_U - g < r_D$ (crossing at $g = 7\%$ with $r_U = 12\%$, $r_D = 5\%$; figure in L03-01-tax-shield-mm-taxes §8). Note that the two compare different policies.

11. *With annual rebalancing the shield is worth more than with continuous rebalancing.* — **T.** The next shield is known one period ahead and discounted at $r_D < r_U$ for that period: factor $(1+r_U)/(1+r_D) > 1$.

12. *The after-tax WACC $\frac{E}{V}r_E + \frac{D}{V}r_D(1-\tau)$ is exact for any debt policy.* — **F.** It is exact under a continuously maintained target ratio (note 02 (5)); with a fixed debt schedule D/V changes and no single WACC exists (slide 32).

13. *When D/E is constant, APV and WACC give the same value.* — **T.** Slides 25–27: substituting $\frac{E}{V}r_E + \frac{D}{V}r_D = r_U$ into the WACC value yields $V_U + \frac{Dr_D\tau}{r_U - g}$.

14. *A fixed 60,000 loan repaid in five years adds more value than a perpetual 25% target ratio, because its annual shield is larger.* — **F.** The loan's shields are 1,200 a year but stop after five years: 5,195 vs. 6,250 for the smaller, growing, perpetual stream.

15. *In APV, the shield of a fixed repayment schedule should be discounted at r_D.* — **T.** Its size is fixed in currency, so its risk is the debt's (slide 21: "use the appropriate r_{ITS}").

16. *For a risky bond, the tax deduction equals the bond's expected return times its market value.* — **F.** The deduction is the coupon actually paid; the slides' example has $r_D = 33.33\%$ on 525 (175) while the interest is the coupon. Use $\bar{r}_D = E[C]/D_t$.

17. *Bankruptcy is an MM violation because ownership passes from shareholders to creditors.* — **F.** The transfer itself is costless in MM (slide 36); the violation is the deadweight cost of the process.

18. *Indirect distress costs can arise before any bankruptcy filing.* — **T.** Slide 38: lost customers, suppliers' and banks' credit, workers, and investment distortions occur as the probability of distress rises (e.g. Hortaçsu et al.'s car prices and CDS).

19. *Debt overhang leads shareholders to take negative-NPV gambles.* — **F.** That is risk-shifting. Debt overhang is underinvestment: shareholders reject positive-NPV projects whose payoff partly goes to creditors (slide 40).

20. *In the slides' model, if the bankruptcy cost is zero, the firm-value-maximizing K is the highest cash flow $\bar{X}$.* — **T.** With $C = 0$, $(1 + r_f)V' = \tau[1 - H(K)] > 0$ for $K < \bar{X}$; at $K = \bar{X}$ the whole tax bill is shielded.

21. *In the slides' model, with no taxes any level of debt leaves firm value unchanged.* — **F.** With $\tau = 0$, $V = V_U - CH(K)/(1 + r_f)$: safe debt ($K \leq \underline{X}$) leaves value unchanged, risky debt destroys $CH(K)/(1 + r_f)$.

22. *At the optimum, the probability of default equals C/τ times the density at K^*.* — **F.** The FOC is $\tau[1 - H(K^*)] = Ch(K^*)$: the **survival** probability is $(C/\tau)h(K^*)$.

23. *Higher capital requirements raise banks' funding costs because equity has a higher required return than debt.* — **F.** Under MM the required return on equity falls as leverage falls and the WACC is unchanged (12% at E/V 25% and 50% in note 05). Only the lost tax subsidy raises the bank's private cost.

24. *If the investment plan is fixed, paying a larger dividend this year financed by a share issue leaves the firm's total value unchanged.* — **T.** Sources = uses forces the issue; substituting it into the pricing equation removes the dividend: $V_t = \delta E[X_{t+1} - I_{t+1} + V_{t+1}]$ (slide 51).

Links

Map: L03-00-map · Formulas: L03-06-formulas · Code: L03-08-python · Grading conventions: rules/00 · Quiz: quiz Topic 2B.

L03 · Python — the lecture's routines, each with self-checks

Prev: L03-07-pitfalls · Next: L03-00-map · Map: L03-00-map

Four blocks, run in order in one session (`python -I`, numpy/scipy only). Each ends with `asserts` against the slides' or the books' own numbers; a broken routine fails loudly. Figures are built separately by `figures/make_figures.py`; the companion's JavaScript is checked against an independent Python copy in `companions/check_companions.py`.

Reused, not rebuilt: the numerical check of the R3 list's solutions, `Listas/lista-r3-mm-taxes-tradeoff-solutions_check.py` (192 checks: tax shields, APV, Hamada, two-state trade-off, Miller's G_L), is run as is in §5. It is a flat script with no importable routines, so the valuation routines below are written fresh.

---

1 · Tax shields, WACC and APV (notes 01–02)

`v_its` implements slide 13's three policies (L03-01-tax-shield-mm-taxes (14)–(16), with the $1 + r_D$ correction); the checks reproduce slides 2–3, 14–15, 28–32 and 63, Ross's firm L and Berk's Example 16.4.

```
import sys
sys.stdout.reconfigure(encoding="utf-8")


def its(tau, r_d, debt):
    """Interest tax shield of one period, slide 3: ITS = tau * r_D * D."""
    return tau * r_d * debt


def v_its(tau, r_d, debt, policy, r_u=None, g=0.0):
    """PV of interest tax shields, slide 13 (note 01 eqs. 14-16).

    policy: "constant" (fixed debt level, r_ITS = r_D), "continuous" (target D/E
 ↪  rebalanced
    continuously, r_ITS = r_U) or "annual" (target D/E reset once a year). r_u, g are
 ↪  needed
    for the two target-ratio policies; raises ValueError if r_u <= g.
    """
    if policy == "constant":
        return tau * debt
    if r_u is None or r_u <= g:
        raise ValueError("target-ratio shields need r_U > g")
    cont = tau * r_d * debt / (r_u - g)
    return cont if policy == "continuous" else cont * (1 + r_u) / (1 + r_d)


def wacc_after_tax(e_v, r_e, r_d, tau):
    """Slide 16: r_WACC = E/V r_E + D/V r_D (1 - tau)."""
    return e_v * r_e + (1 - e_v) * r_d * (1 - tau)
```

```
def annuity(r, n):
    """PV of 1 per period for n periods at rate r."""
    return (1 - (1 + r) ** -n) / r


# Slides 2-3 (Berk Table 16.2): taxes 525 vs 647.5, ITS 122.5
assert abs(0.35 * 1850 - 0.35 * 1500 - its(0.35, 1.0, 350)) < 1e-9
# Slides 14-15: V_L = 800, P = 0.8, N = 625, gain 200
vl = 600 + v_its(0.40, 0.05, 500, "constant")
P = vl / 1000
assert abs(vl - 800) < 1e-9 and abs(500 / P - 625) < 1e-9 and abs((P - 0.6) * 1000 -
 ↪  200) < 1e-9
# Ross pp. 551-553: V_L = 7,300, R_E = 10.22%, WACC 9.589%
E = 7000 + 0.30 * 1000 - 1000
r_e = 0.10 + (0.10 - 0.08) * 1000 / E * (1 - 0.30)
assert abs(E - 6300) < 1e-9 and round(100 * r_e, 2) == 10.22
assert abs(wacc_after_tax(E / 7300, r_e, 0.08, 0.30) - 0.10 * (1 - 0.30 * 1000 /
 ↪  7300)) < 1e-12
# Slides 28-31: WACC 11.5%, V_L = 100,000 both ways; slide 32: 98,945
r_u, r_d, tau, g = 0.12, 0.05, 0.40, 0.04
r_e = r_u + (0.25 / 0.75) * (r_u - r_d)                  # note 01 (13): 14.333%
w = wacc_after_tax(0.75, r_e, r_d, tau)
assert abs(w - 0.115) < 1e-12 and abs(w - (r_u - tau * r_d * 0.25)) < 1e-12   # note
 ↪  02 (1) = (4)
vu = 7500 / (r_u - g)
assert abs(7500 / (w - g) - 100000) < 1e-6 and abs(vu - 93750) < 1e-6
assert abs(vu + v_its(tau, r_d, 25000, "continuous", r_u, g) - 100000) < 1e-6
assert abs(v_its(tau, r_d, 25000, "annual", r_u, g) - 6666.6667) < 1e-3
loan = its(tau, r_d, 60000) * annuity(r_d, 5)
assert round(loan) == 5195 and round(vu + loan) == 98945
# Berk Example 16.4 (p. 476): 42 million a year for 10 years at 6% -> 309 million
assert round(42 * annuity(0.06, 10)) == 309
# Slide 63: r_D = 33.33% but the shield follows the coupon rate 50/525
r_D, rbar = 700 / 525 - 1, 50 / 525
assert abs(r_D - 1 / 3) < 1e-12 and abs((r_D - 0.35 * rbar) - 0.30) < 1e-12
print("section 1 OK")
```

2 · The appendix identity, numerically

Section 1 already checks that the after-tax WACC (note 02 (1)) equals $r_U - \tau r_D D/V$ (note 02 (4)) and that discounting at it gives the APV value — the content of the appendix derivation (note 02 (14)–(20)) in the constant-ratio case.

3 · The static trade-off with any density (note 04)

`TradeOff` integrates slides 42–44 numerically for any h, so it does **not** rely on the uniform closed forms used by the figures and the companion — it is the independent path. The checks: $K^* = \bar{X} - C/\tau$ and $V(K^*) = 81.6327$ for the uniform calibration, $E + D = V$, identity (8), FOC root = grid argmax, the two landmarks, a symmetric triangular density with the signs of (13)–(14), and the volatility caveat of

note 04 §6.

```
import numpy as np
from scipy import integrate, optimize, stats


class TradeOff:
    """Slides 41-45: one-period trade-off with any density h on [x_lo, x_hi].

    pdf, cdf: callables for h and H; tau: tax rate paid only if solvent; c: fixed
    default
    cost (slide assumes c < x_lo); rf: investors' required return. All values are at
    t = 0.
    """

    def __init__(self, pdf, cdf, x_lo, x_hi, tau, c, rf):
        self.h, self.H, self.lo, self.hi = pdf, cdf, x_lo, x_hi
        self.tau, self.c, self.rf = tau, c, rf

    def _int(self, f, a, b):
        a, b = max(a, self.lo), min(b, self.hi)
        return integrate.quad(lambda x: f(x) * self.h(x), a, b)[0] if b > a else 0.0

    def v_u(self):                          # eq. (2)
        return (1 - self.tau) * self._int(lambda x: x, self.lo, self.hi) / (1 +
         self.rf)

    def equity(self, K):                    # eq. (4)
        return (1 - self.tau) * self._int(lambda x: x - K, K, self.hi) / (1 +
         self.rf)

    def debt(self, K):                      # eq. (5)
        return (self._int(lambda x: x - self.c, self.lo, K)
                + K * (1 - self.H(min(max(K, self.lo), self.hi)))) / (1 + self.rf)

    def value(self, K):                     # eq. (6)
        Kc = min(max(K, self.lo), self.hi)
        s = self._int(lambda x: x, self.lo, K) + K * (1 - self.H(Kc))
        return self.v_u() + (self.tau * s - self.c * self.H(Kc)) / (1 + self.rf)

    def foc(self, K):                       # eq. (10) times (1 + rf)
        return self.tau * (1 - self.H(K)) - self.c * self.h(K)

    def k_star(self):
        """Root of (11) on (x_lo, x_hi); corner x_lo if the FOC is already negative
         there."""
        a, b = self.lo + 1e-9, self.hi - 1e-9
        if self.c == 0:
            return self.hi
        if self.foc(a) <= 0:
```

```
            return self.lo
        return optimize.brentq(self.foc, a, b)


def uniform(lo=50.0, hi=150.0, tau=0.35, c=20.0, rf=0.05):
    u = stats.uniform(loc=lo, scale=hi - lo)
    return TradeOff(u.pdf, u.cdf, lo, hi, tau, c, rf)


m = uniform()
ks = m.k_star()
assert abs(ks - (150 - 20 / 0.35)) < 1e-7                                    # eq. (16)
assert abs(m.value(ks) - 81.6327) < 1e-4 and abs(m.v_u() - 61.9048) < 1e-4
assert abs(m.debt(ks) - 71.5258) < 1e-4 and abs(m.equity(ks) - 10.1069) < 1e-4
assert abs(m.debt(ks) + m.equity(ks) - m.value(ks)) < 1e-9                   # E + D =
 ↪  V
assert abs(m.value(ks) - (m.v_u() + m.tau * m.debt(ks)
                          - (1 - m.tau) * m.c * m.H(ks) / (1 + m.rf))) < 1e-9   # eq.
                           ↪  (8)
grid = np.linspace(0, 180, 1801)
assert abs(grid[np.argmax([m.value(k) for k in grid])] - ks) < 0.1          # FOC =
 ↪  grid argmax
assert abs(m.value(50) - 78.5714) < 1e-4 and abs(m.value(160) - 76.1905) < 1e-4
assert uniform(c=0).k_star() == 150 and uniform(c=40).k_star() == 50        #
 ↪  landmarks / corner

# A non-uniform h: symmetric triangular on [50, 150]; comparative statics (13)-(14)
 ↪  numerically
def tri(tau=0.35, c=20.0):
    t = stats.triang(c=0.5, loc=50, scale=100)
    return TradeOff(t.pdf, t.cdf, 50, 150, tau, c, 0.05)

k0 = tri().k_star()
assert 50 < k0 < 150 and abs(tri().foc(k0)) < 1e-9
assert tri(tau=0.40).k_star() > k0 > tri(c=25.0).k_star()                    # dK*/dtau
 ↪  > 0, dK*/dC < 0

# Note 04 section 6 caveat: mean 100, widths 20, 100, 140 (C < X_lo holds) -> value
 ↪  gains 30.00, 19.73, 18.17
for w_, gain in [(20, 30.00), (100, 19.73), (140, 18.17)]:
    mw = uniform(lo=100 - w_ / 2, hi=100 + w_ / 2)
    assert round(mw.value(mw.k_star()) - mw.v_u(), 2) == gain
print("section 3 OK: K* uniform", round(ks, 4), "triangular", round(k0, 4))
```

4 · Dividend irrelevance (note 05)

Prices per share (slide 50, (1)) and share counts (sources = uses, (2)) depend on each other, so the routine solves them jointly; the value $n_0 P_0$ comes out equal to (6) for five random payout plans with the investment plan fixed.

```python
import numpy as np

rng = np.random.default_rng(0)


def firm_value_by_payout(X, I, div, delta, n0=1.0, iters=500):
    """Slides 49-51, deterministic: V_0 = n_0 P_0 for a given dividend plan.

    X, I, div: arrays for t = 1..T of cash flow, investment and TOTAL dividends paid to the
    shares outstanding at t-1; the gap div_t + I_t - X_t is raised by issuing (or bought back
    by repurchasing) shares at the ex-dividend price P_t, eq. (2). The firm is liquidated at T,
    so P_T = 0 and div_T must equal X_T - I_T. Prices P_t = delta (d_{t+1} + P_{t+1}), eq. (1),
    and share counts depend on each other, so they are solved by fixed-point iteration.
    """
    T = len(X)
    assert abs(div[-1] - (X[-1] - I[-1])) < 1e-12, "nothing can be raised at P_T = 0"
    raised = div + I - X                        # index k <-> period k+1
    P = np.ones(T + 1)
    P[T] = 0.0
    n = np.full(T + 1, n0)
    for _ in range(iters):
        for t in range(1, T):                   # n_t = n_{t-1} + raised_t / P_t
            n[t] = n[t - 1] + raised[t - 1] / P[t]
        n[T] = n[T - 1]
        for t in range(T - 1, -1, -1):          # eq. (1), per-share dividend d_{t+1} = div / n_t
            P[t] = delta * (div[t] / n[t] + P[t + 1])
    return n[0] * P[0]


X = np.array([10.0, 12, 9, 15, 11, 14])
I = np.array([4.0, 6, 2, 8, 3, 0])
delta = 1 / 1.08
target = sum(delta ** (i + 1) * (X[i] - I[i]) for i in range(len(X)))       # eq. (6)
for _ in range(5):
    div = rng.uniform(0.6, 1.4, len(X)) * (X - I)
    div[-1] = X[-1] - I[-1]
    assert abs(firm_value_by_payout(X, I, div, delta) - target) < 1e-9
# Ross Wharton (p. 588): 173.55 under both policies
assert round(100 / 1.1 + 100 / 1.21, 2) == round(110 / 1.1 + 89 / 1.21, 2) == 173.55
# Berk Genron (pp. 503-507): 42 under the three policies
assert abs(2 + 4.80 / 0.12 - 42) < 1e-9 and abs(48 / (10 - 20 / 42) / 0.12 - 42) < 1e-9
assert abs(48 / (10 + 28 / 42) * (1 + 1 / 0.12) - 42) < 1e-9
print("section 4 OK: V_0 =", round(target, 4), "for every payout policy")
```

5 · The R3 list's own check (reuse)

```
import runpy
# run from the repository root
runpy.run_path("Listas/lista-r3-mm-taxes-tradeoff-solutions_check.py")    # prints
 ↪  "PASS 192 checks"
```

Links

Map: L03-00-map · Formulas: L03-06-formulas · Pitfalls: L03-07-pitfalls · Figures: `figures/make_figures.py` · Companion check: `companions/check_companions.py` · List R3: Lista R3.